University of Hohenheim

Time Series Approaches to Cocoa Price Risk

Master Thesis

submitted in partial fulfillment of the requirements for the degree of

Master of Agricultural Sciences – Advisory and Innovation Services in Agri-Food Systems

Author:	Felix Köhler
First Supervisor:	Prof. Dr. Thomas Dimpfl
Second Supervisor:	Dr. Johannes Bleher

Contents

1 Introduction

Volatility forecasting is fundamental to financial risk management, portfolio allocation, and derivative pricing (Poon and Granger 2003). Two modeling paradigms dominate the literature. The first treats volatility as latent and inferred from returns; the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model of Bollerslev (1986) is the canonical approach. The second, enabled by high-frequency data, treats volatility as directly observable through realized measures, and the Heterogeneous Autoregressive (HAR) model of Corsi (2009) has become the dominant forecasting framework within this paradigm, capturing long-memory properties of volatility in a parsimonious, OLS-estimable structure.

The majority of volatility forecasting research focuses on conditional mean forecasts: the expected level of future volatility given current information. Yet the conditional mean provides an incomplete characterization of future volatility. When the volatility distribution is skewed or fat-tailed, as is typical for commodity markets, the conditional quantiles can diverge substantially from the mean, and understanding how the entire distribution shifts with predictor values becomes important. Quantile regression, introduced by Koenker and Bassett (1978), provides a natural framework for estimating conditional quantiles directly, without distributional assumptions.

Cocoa futures present a particularly relevant case for distributional volatility forecasting. Production is geographically concentrated in West Africa, biologically slow to adjust, and exposed to both disease and climate shocks. Between 2022 and 2024, international cocoa prices approximately tripled, an increase unprecedented in the commodity's modern trading history. Despite this salience, dedicated cocoa volatility research remains limited, and the application of HAR-based quantile regression to cocoa, or to any agricultural commodity, is an open gap in the literature. At long forecast horizons, where the autoregressive signal from lagged volatility fades, climate variables offer an additional source of predictive information. The El Niño–Southern Oscillation (ENSO) influences West African rainfall patterns with effects on cocoa yields hypothesised to materialise 6–12 months after the climate signal. Evidence from oil and broad agricultural commodity surveys suggests that ENSO can improve volatility forecasts at these longer horizons (Bouri et al. 2021; Bonato et al. 2023), motivating a test of whether this channel extends to cocoa.

This thesis addresses two research questions:

RQ1 Can HAR-based quantile regression models forecast cocoa futures volatility better than baseline models (historical volatility, GARCH, HAR-OLS) at horizons from 1 day to 12 months?

RQ2 Does the ENSO climate index improve volatility forecasts, especially at long horizons?

We address these questions by estimating HAR-QR models at multiple quantiles ($\tau \in \{0.50, 0.75, 0.95\}$) and horizons (1 day, 1 week, 1 month, 3 months, 6 months, 12 months). At each horizon, we compare out-of-sample forecast accuracy against standard benchmarks: historical rolling volatility, GARCH(1,1), and HAR estimated by ordinary least squares. For long horizons, we test whether adding ENSO improves forecasts, with particular focus on the upper quantile ($\tau = 0.95$).

The thesis makes three contributions. First, it provides the first application of HAR-QR to an agricultural commodity, extending a framework previously confined to equity and precious-metals markets. Second, it tests HAR-QR at horizons of 6–12 months, far beyond the current literature limit of approximately one month (Lyócsa and Stašek 2021). Third, it tests whether ENSO enhances quantile volatility forecasts for cocoa, a combination untested in the literature.

The remainder of the thesis is organized as follows. Section 2 reviews the literature on volatility measurement, HAR models, quantile regression in finance, and climate-commodity linkages, establishing the research gap this thesis addresses. Section 3 specifies the HAR-QR model, the multi-horizon projection approach, and the evaluation framework. Section 4 describes the ICE London cocoa futures data and validates the daily volatility proxy against realized volatility from intraday data. Section 5 presents empirical results organized by forecast horizon. Section 6 interprets findings, discusses limitations, and considers practical implications.

2 Literature Review

This section reviews the literature underlying volatility forecasting for cocoa futures. We begin with the cocoa market itself, then trace how volatility measurement evolved from simple estimators to the HAR framework. We then review quantile regression for volatility and ENSO as a climate predictor at long horizons, followed by the evaluation methods used to compare forecasts. The intersection of these strands identifies the gap this thesis addresses: no study applies HAR-based quantile regression to any agricultural commodity, and no HAR-QR study extends beyond approximately one month.

2.1 The Cocoa Market

Cocoa occupies an unusual position among agricultural commodities. Its production is geographically concentrated, biologically slow to adjust, and structurally exposed to climate variability in ways that set it apart from grains or oilseeds. These features interact to produce a market where supply disruptions are both frequent and slow to resolve, precisely the conditions under which distributional, rather than mean, volatility forecasts become most valuable.

The West African cocoa belt accounts for approximately 70% of world production, with Côte d'Ivoire and Ghana alone responsible for more than half of global output (Schroth et al. 2016; Läderach et al. 2013). Unlike grains or oilseeds, whose production is spread across multiple continents, cocoa's concentration means that disruptions in a single region can move global prices substantially. Production is overwhelmingly smallholder-based, with approximately 90% grown on farms smaller than five hectares (Bensch et al. 2022). Since the mid-2000s, growing financial participation has added a further source of volatility (Bensch et al. 2022; Gilbert and Morgan 2010; Rogna and Tillie 2025).

A critical structural feature is the cocoa tree's biological rigidity. New trees require five to seven years before reaching full productive capacity, with an economic lifespan of 30 to 40 years (Asante et al. 2025). Unlike annual crops, cocoa supply responds to market signals slowly, and the resulting low short-run supply elasticity amplifies price volatility whenever demand or supply shocks occur.

Cocoa is also among the most climate-sensitive perennial crops. West Africa is characterized by variable climates (Läderach et al. 2013), and drought during the critical September–December pod development period reduces yields substantially. The El Niño–Southern Oscillation (ENSO) is the primary near-term mechanism through which climate variability affects cocoa supply; Section 2.5 discusses this channel and its implications for long-horizon volatility forecasting.

The most striking recent illustration of these compounding dynamics is the 2022–

2024 price crisis, during which international cocoa prices approximately tripled (Rogna and Tillie 2025). The crisis was driven by the convergence of the Cocoa Swollen Shoot Virus Disease (CSSVD) in Ghana and adverse El Niño–related drought across Côte d'Ivoire.

Despite the salience of cocoa in global commodity markets, the volatility forecasting literature has paid it limited attention. Bensch et al. (2022) provide the most comprehensive study of cocoa price determinants, finding that fundamentals are quickly integrated into prices, but do not extend to multi-horizon forecasting or tail-risk quantification. Bergander (2025) apply LSTM neural networks with weather data but find no improvement in point forecasts, plausibly because markets anticipate weather shocks before physical effects materialize. Within broader commodity surveys, Alfeus and Nikitopoulos (2022) include cocoa among 22 markets and find that monthly volatility is the most informative component, evidence of slow-moving structural dynamics, but offer no focused analysis. No study applies HAR-based models specifically to cocoa volatility, and none has examined quantile-based tail-risk forecasting for this market.

2.2 The Evolution of Volatility Measurement

A foundational challenge in volatility research is that volatility is latent: unlike prices, which are directly observable, it must be estimated from price movements (Poon and Granger 2003). Every volatility measure is an approximation, and the choice of estimator shapes both forecast quality and evaluation validity (Patton 2011). The measurement literature has evolved through three successive approaches, each reducing the statistical noise of the volatility proxy.

The earliest volatility estimates used only closing prices. The random walk model, exponential smoothing, and rolling-window standard deviations all extrapolate past variation into the future (Poon and Granger 2003). These methods are simple and require minimal data, but the squared close-to-close return is a highly imprecise estimator of daily variance. Nonetheless, Poon and Granger (2003) review 93 studies and find that "for forecast horizons that are longer than 6 months, a simple historical method using low frequency data over a period at least as long as the forecast horizon works best," a finding that motivates the inclusion of rolling historical volatility as a benchmark in this thesis.

A significant efficiency improvement came from exploiting the full daily price range. Parkinson (1980) demonstrated that the high-low range achieves efficiency gains of approximately five times over close-to-close methods. Rogers and Satchell (1991) extended this to allow for non-zero drift. However, both estimators assume continuous trading, an assumption violated by commodity futures markets, which experience regular overnight closures.

Yang and Zhang (2000) synthesized these approaches into an estimator that is unbiased, drift-independent, and robust to opening jumps. By estimating overnight and trading-session variance separately and combining them optimally, the Yang-Zhang estimator achieves efficiency gains of 7–8 times over close-to-close methods. For commodity markets, where overnight information frequently causes opening gaps, this robustness is particularly valuable.

The availability of intraday transaction data in the late 1990s enabled a fundamentally different approach. Andersen and Bollerslev (1998) and Andersen et al. (2001) demonstrated that realized volatility (the sum of squared intraday returns) provides a far more precise measure of daily variance, resolving much of the apparent poor performance of earlier models. However, high-frequency data are plagued by microstructure noise and are often unavailable for commodity markets over long samples. Clements et al. (2024) provide evidence that HAR models applied to daily OHLC-based volatility achieve comparable forecasting performance at medium and long horizons, validating the use of daily data when intraday records are limited, as is the case for cocoa futures over multi-decade samples.

These developments establish the Yang-Zhang estimator applied to daily OHLC data as the volatility proxy for this thesis. The following section turns from the question of what to measure to the question of how to model and forecast its temporal dynamics.

2.3 The Heterogeneous Autoregressive Model

Two modeling frameworks are central to this thesis: GARCH, which serves as the main benchmark, and the Heterogeneous Autoregressive (HAR) model, which is the primary forecasting framework. Both capture a defining feature of volatility that range-based estimators ignore: its tendency to cluster over time.

Volatility clustering was first formalized by Engle (1982), and Bollerslev (1986) generalized this into the GARCH framework, capturing both clustering and mean reversion. The GARCH(1,1) has proven remarkably difficult to beat: Hansen and Lunde (2005) evaluate 330 volatility models and conclude that "nothing beats a GARCH(1,1)," establishing it as the natural benchmark. However, GARCH cannot replicate the slow, hyperbolic decay of volatility autocorrelations observed empirically, a property known as long memory. As Corsi (2009) observes, "standard GARCH and stochastic volatility short-memory models appear like white noise once aggregated over longer time periods."

The HAR model, introduced by Corsi (2009), addresses this limitation through a parsimonious structure that captures long memory without the estimation difficulties of fractionally integrated models. It builds on the *Heterogeneous Market Hypothesis* of Müller et al. (1997), who documented that coarse-grained volatility (e.g., weekly) predicts fine-grained volatility (e.g., daily) more strongly than the reverse. This asymmetry reflects the presence of traders with different investment horizons: intraday speculators, swing traders, and long-term institutional investors, whose actions generate volatility that cascades across frequencies. Corsi formalized this insight into an additive model with daily, weekly, and monthly volatility components: a restricted AR(22) estimable by OLS that reproduces the autocorrelation patterns of long-memory processes.

Since its introduction, the HAR framework has been extended with jump components, realized semivariance, and time-varying coefficients (Andersen et al. 2007; Patton and Sheppard 2015; Tian et al. 2017). However, for agricultural commodities, Degiannakis et al. (2022) find that "sophisticated HAR-type models are not capable of outperforming the simple HAR in an out-of-sample exercise." This sets a high bar for any augmentation to clear, and motivates subjecting any extension to rigorous out-of-sample testing before drawing conclusions.

For cocoa specifically, Alfeus and Nikitopoulos (2022) include it among 22 commodities in a broad HAR study, finding that "monthly volatility matters the most" for cocoa, evidence that longer-term participants exert particular influence on commodity volatility dynamics. Haugom et al. (2016) provide complementary evidence that "the effect from the monthly volatility component is strongest for all assets when predicting conditional tails." Despite this, no study focuses specifically on cocoa volatility dynamics or applies HAR to horizons beyond one month.

2.4 Quantile Regression for Volatility

Standard volatility models, whether GARCH or HAR, estimate the conditional mean of volatility. Yet when the underlying process exhibits fat tails and regime shifts, the conditional quantiles can diverge substantially from the mean. Quantile regression, introduced by Koenker and Bassett (1978), estimates conditional quantiles directly, with each quantile having its own parameter vector. This allows the relationship between predictors and the dependent variable to differ across the distribution without imposing parametric assumptions on the error distribution.

Baur and Dimpfl (2019) demonstrate that quantile autoregression encompasses both the conditional mean and conditional variance of the return distribution through the inter-quantile range. Their approach is robust to jumps and produces variance forecasts comparable to, and in some cases more accurate than, GARCH and HAR alternatives. Haugom et al. (2016) introduced HAR-QREG, combining the multi-horizon HAR structure with quantile regression, and show that the model passes conditional coverage tests at substantially higher rates than historical simulation.

HAR-QR has since been applied in equity and precious-metals markets. Lyócsa and

Stašek (2021) develop a complete subset quantile regression variant achieving significant improvements at horizons up to 22 trading days; Huang et al. (2022) confirm the model's superiority using high-frequency data for Chinese indices; and Song et al. (2025) show that HAR-QR parameters improve machine learning forecasts for gold.

Despite this growing body of work, two gaps remain. First, no study applies HAR-QR to any agricultural commodity (Degiannakis et al. 2022; Bonato et al. 2023). Second, existing studies are confined to short horizons: Lyócsa and Stašek (2021) extend to 22 trading days, the longest in the literature. Whether quantile forecasts retain predictive value at horizons of 6–12 months remains untested.

2.5 Climate Variables and Commodity Volatility

At long forecast horizons, the autoregressive signal from lagged volatility fades. Climate variables offer a potential source of predictive information that operates on precisely these longer timescales.

The El Niño–Southern Oscillation (ENSO) is an irregularly periodic fluctuation in Pacific sea surface temperatures that affects weather patterns globally, cycling between warming (El Niño) and cooling (La Niña) phases every two to seven years. For cocoa, the transmission mechanism is direct: during El Niño phases, West Africa typically receives below-normal rainfall during the growing season, reducing yields in subsequent harvests. Schroth et al. (2016) document that "drought years regularly affect cocoa yields in West Africa and have particularly done so during severe El Niño years." With the main crop harvested from October through March and the mid-crop from May through August, ENSO-driven rainfall anomalies are hypothesised to translate into production changes six to twelve months after the climate signal. We refer to this chain, from climate signal through rainfall deficit to harvest disruption and price volatility, as the *agronomic-transmission hypothesis*. The implied transmission window, if confirmed empirically, aligns with the long forecast horizons this thesis examines. Importantly, markets respond not only to realized weather but to climate expectations: Bergander (2025) observe that roughly 85% of the eventual price impact of a drought shock is reflected in futures prices before the actual crop loss occurs. This forward-looking price adjustment does not preclude predictability in *volatility*: even when the expected future price level rapidly impounds climate information, the conditional dispersion of realized outcomes may remain systematically forecastable from the ENSO state.

At the commodity price level, Ubilava (2018) examines ENSO–price relationships across 43 commodities, finding significant effects for tropically grown agricultural products including cocoa, with "more amplified price responses during El Niño events." Recent work has extended this analysis from price levels to volatility. Bouri et al. (2021) find that ENSO improves oil volatility forecasts at horizons of 2–4 years, far beyond where lagged volatility alone provides value. Bonato et al. (2023) extend this to 16 agricultural commodities, finding that "the evidence of predictive value of El Niño and La Niña events strengthens at the longer term forecast horizons," precisely where standard autoregressive models lose power. More broadly, Guo et al. (2025) demonstrate that climate risk indices significantly improve commodity volatility forecasts, with models incorporating climate risk "outperforming traditional ones in economic value."

These findings establish ENSO as a physically motivated, exogenous predictor with demonstrated value for oil and agricultural commodity volatility at long horizons, but whether this extends to cocoa remains untested, and no prior study has combined ENSO with quantile regression.

2.6 Evaluating Volatility Forecasts

A distinctive challenge in evaluating volatility forecasts is that the variable of interest is unobservable, even ex post (Patton 2011). Every volatility measure is an imperfect proxy, and noise in the proxy can produce misleading model rankings depending on the loss function used.

Patton (2011) derives conditions for loss functions that yield rankings invariant to the choice of volatility proxy. Among common loss functions, only mean squared error (MSE) and QLIKE satisfy these conditions. Non-robust loss functions can produce rankings that deviate by factors of one quarter to three, explaining widespread conflicting findings across studies. Between the two robust measures, QLIKE penalizes under-prediction proportionally more heavily, making it suitable when the forecasting goal involves risk assessment (Lyócsa and Stašek 2021; Clements et al. 2024; Ghysels and Marcellino 2018). Mean absolute error (MAE) is a further robustness check that captures deviations in the volatility level rather than its square.

Hansen et al. (2011) propose the Model Confidence Set (MCS) for simultaneous comparison of multiple models, identifying the set $\mathcal{M}^*$ containing the best model at a given confidence level. The MCS "acknowledges the limitations of the data": informative samples yield small confidence sets, while uninformative samples include many models, particularly relevant when evaluation noise from an unobservable target reduces the power to discriminate (Clements et al. 2024; Degiannakis et al. 2022).

For quantile forecasts, Christoffersen (1998) establishes conditional coverage testing: a well-specified quantile forecast at level τ should produce violations at rate τ, and these violations should be serially independent. The quantile loss function (the check function used in estimation) also serves as a natural evaluation metric, penalizing over- and under-prediction asymmetrically according to τ (Lyócsa and Stašek 2021). Puke and Schweikert (2026) show that aligning the estimation and evaluation loss function produces significant forecast gains; since HAR-QR uses the check function for both estimation and evaluation, it is inherently coherent by construction.

3 Methodology

This section specifies the econometric framework for forecasting cocoa futures volatility at horizons from one day to twelve months. We first define the volatility measure, then specify the HAR-QR model and its ENSO-augmented extension, describe the benchmark models, and finally lay out the evaluation framework. Table 1 provides an at-a-glance summary of the five models before the detailed specifications follow.

Table 1: Overview of models compared in this study, ordered from simplest to most complex.

	Benchmarks			Main models	
Feature	Hist. Vol.	GARCH(1,1)	HAR-OLS	HAR-QR	HAR-X-QR
Model class	Non-param.	ARCH-type	Linear reg.	Quantile reg.	Quantile reg.
HAR lags ($d/w/m$)	No	No	Yes	Yes	Yes
ENSO regressor	No	No	No	No	Yes
Quantile output	Empirical	Simulation	None	Direct	Direct
Estimation	—	MLE	OLS	Check fn.	Check fn.
Multi-period design	Rolling window	Recursive	Direct	Direct	Direct
Thesis role	Naive BM	Param. BM	HAR benchmark	Main (RQ1)	Main (RQ2)

Direct = horizon-specific model estimated by direct projection; Recursive = closed-form GARCH mean recursion, quantiles via Gaussian closed-form (Eq. 13); Check fn. = check-function (quantile) loss; BM = benchmark; Hist. Vol. = rolling 252-day window with empirical quantiles.

3.1 Volatility Measure

We measure daily volatility using the Yang-Zhang estimator, which combines overnight and trading-session variance components:

$$\hat{\sigma}^2_{\mathrm{YZ}} = \hat{\sigma}^2_O + k\hat{\sigma}^2_C + (1-k)\hat{\sigma}^2_{\mathrm{RS}}, \tag{1}$$

where $\hat{\sigma}^2_O$ captures overnight variance, $\hat{\sigma}^2_C$ captures open-to-close variance, and $\hat{\sigma}^2_{\mathrm{RS}}$ is the Rogers-Satchell component that exploits the daily range. The weighting factor $k = 0.34/(1.34 + (n+1)/(n-1))$ minimizes estimator variance.

This choice is motivated by two considerations. First, Yang-Zhang achieves efficiency gains over close-to-close estimators and second, it can be computed over our full 40-year daily sample, unlike realized volatility which requires high-frequency data. We validate the Yang-Zhang proxy against five-minute realized volatility in Section 4, comparing rolling-window aggregates (22-day averages) of both measures via a Mincer-Zarnowitz regression. This comparison at the multi-day level aligns with the HAR model's use of averaged volatility as both predictors and targets.

3.2 The HAR-QR Model

Combining the HAR framework with quantile regression yields the HAR-QR model. The HAR-QR model for horizon h and quantile τ is

$$Q_\tau(\bar{\sigma}_{t:t+h} \mid \mathcal{F}_t) = \beta_0(\tau) + \beta_d(\tau)\sigma_t^{(d)} + \beta_w(\tau)\sigma_t^{(w)} + \beta_m(\tau)\sigma_t^{(m)}, \tag{2}$$

where $\mathcal{F}_t$ denotes the information set at time t. The volatility components capture variation at three frequencies:

$$\sigma_t^{(d)} = \sigma_t, \tag{3}$$

$$\sigma_t^{(w)} = \frac{1}{5}\sum_{i=0}^{4} \sigma_{t-i}, \tag{4}$$

$$\sigma_t^{(m)} = \frac{1}{22}\sum_{i=0}^{21} \sigma_{t-i}. \tag{5}$$

The key feature of quantile regression is that each quantile τ has its own coefficient vector, allowing the relationship between predictors and future volatility to differ across the distribution. For the median ($\tau = 0.50$), the coefficients may resemble those from OLS estimation. For the upper quantile ($\tau = 0.95$), the coefficients capture how the predictors relate to volatility during high-volatility episodes. We report three quantiles: $\tau \in \{0.50, 0.75, 0.95\}$. The median $\tau = 0.50$ serves as an anchor to OLS-based point forecasts. The intermediate quantile $\tau = 0.75$ traces the transition from central to tail behaviour. The upper quantile $\tau = 0.95$ is of primary interest, as it captures the conditional upper bound of volatility, the regime where mean forecasts are most likely to understate actual outcomes. Lower quantiles ($\tau = 0.05,\ 0.25$) are omitted as the thesis focuses on upper-tail risk.

The dependent variable is the average daily volatility over the forecast horizon, following the HAR literature convention (Ghysels and Marcellino 2018; Clements et al. 2024):

$$\bar{\sigma}_{t:t+h} = \frac{1}{h}\sum_{i=1}^{h} \sigma_{t+i}. \tag{6}$$

This averaging ensures comparability across horizons and aligns with the HAR model's construction, where weekly and monthly components are themselves averages of daily volatility.

We adopt the *direct* forecasting approach: a separate model is estimated for each horizon h, using $\bar{\sigma}_{t:t+h}$ as the dependent variable. This contrasts with *iterated* forecasting, which estimates a one-step model and iterates forward. Ghysels and Marcellino (2018) find that direct forecasting is preferred for realized volatility models. For quantile regression, direct estimation is particularly natural, as it produces the conditional quantile at horizon h without the distributional assumptions required to iterate quantile forecasts. Our longest horizons (6–12 months) would require iterating hundreds of daily steps, amplifying any model misspecification.

We estimate HAR-QR at six horizons: $h \in \{1, 5, 22, 66, 132, 264\}$ trading days, corresponding to 1 day, 1 week, 1 month, 3 months, 6 months, and 12 months. The short horizons (1 day to 1 month) serve to validate the methodology against existing literature: HAR-QR has been tested at horizons up to 22 trading days (Lyócsa and Stašek 2021). The long horizons (6 and 12 months) constitute the core contribution, testing whether HAR-QR can produce useful forecasts at horizons where no prior study has ventured.

At long horizons, the autoregressive signal from lagged volatility fades. We therefore augment the HAR-QR model with the ENSO climate index:

$$Q_\tau(\bar{\sigma}_{t:t+h} \mid \mathcal{F}_t) = \beta_0(\tau) + \beta_d(\tau)\sigma_t^{(d)} + \beta_w(\tau)\sigma_t^{(w)} + \beta_m(\tau)\sigma_t^{(m)} + \gamma(\tau)\mathrm{ENSO}_t, \tag{7}$$

We refer to this augmented specification as the *HAR-X-QR* model, where the "-X" suffix denotes the inclusion of the exogenous ENSO regressor. Here ENSO_t is the Niño 3.4 sea surface temperature anomaly index from NOAA, measured monthly. The coefficient $\gamma(\tau)$ captures the marginal effect of ENSO conditions on future volatility at quantile τ.

We use *contemporary* ENSO, meaning ENSO_t predicts $\bar{\sigma}_{t:t+h}$, rather than explicitly lagged ENSO values. This specification is standard in the climate-commodity literature (Bouri et al. 2021; Bonato et al. 2023) and is causal: the Niño 3.4 anomaly for month t is publicly observable before any forecast is issued, so no forward-looking information enters the model. The sensitivity of this specification to alternative ENSO lags is examined in Section 5.5.

Positive Niño 3.4 anomalies indicate El Niño conditions; negative anomalies indicate La Niña. Full details of the index construction, sample coverage, and monthly-to-daily alignment are provided in Section 4.2.

The HAR-QR model is estimated by minimizing the check function loss:

$$\hat{\beta}(\tau) = \arg\min_{\beta} \sum_{t=1}^{T-h} \rho_\tau(\bar{\sigma}_{t:t+h} - X_t'\beta), \tag{8}$$

where $X_t = (1, \sigma_t^{(d)}, \sigma_t^{(w)}, \sigma_t^{(m)})'$ for the basic model, or $X_t = (1, \sigma_t^{(d)}, \sigma_t^{(w)}, \sigma_t^{(m)}, \mathrm{ENSO}_t)'$ for the extended model. The check function is defined as:

$$\rho_\tau(u) = u \cdot (\tau - \mathbf{1}_{u<0}) = \begin{cases} \tau \cdot u & \text{if } u \geq 0, \\ (\tau - 1) \cdot u & \text{if } u < 0. \end{cases} \tag{9}$$

This asymmetric loss function penalizes under-prediction more heavily when τ is large and over-prediction more heavily when τ is small, ensuring that the estimated quantile achieves the target coverage rate. The minimization is a linear programming problem solved efficiently using the simplex algorithm (Koenker and d'Orey 1987). Standard errors are computed using the bootstrap.

3.3 Benchmark Models

We compare HAR-QR and HAR-X-QR against three benchmarks that span the major modeling paradigms discussed in the literature review.

The first benchmark is *rolling historical volatility* with a one-year window ($w = 252$):

$$\hat{\sigma}_{t+h}^{\text{hist}} = \sqrt{\frac{1}{w}\sum_{i=1}^{w}(r_{t-i+1} - \bar{r})^2}. \tag{10}$$

For quantile forecasts, we compute empirical quantiles of the historical volatility distribution: sort the w past volatilities and select the value at position $\lceil \tau \cdot w \rceil$. This model captures only unconditional variation and serves as a naive baseline.

The second benchmark is the *GARCH(1,1)* model (Bollerslev 1986):

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2. \tag{11}$$

For mean forecasts at horizon h, the GARCH(1,1) produces closed-form predictions:

$$\mathbb{E}_t[\sigma_{t+h}^2] = \bar{\sigma}^2 + (\alpha+\beta)^{h-1}(\sigma_{t+1}^2 - \bar{\sigma}^2), \tag{12}$$

where $\bar{\sigma}^2 = \omega/(1-\alpha-\beta)$ is the unconditional variance and $(\alpha+\beta)^{h-1}$ governs the speed of mean reversion. At long horizons, all h-step forecasts converge to $\bar{\sigma}^2$, irrespective of current conditions, a structural limitation of daily GARCH at 6–12 month horizons.

For quantile forecasts we adopt a closed-form Gaussian approach consistent with the normal-innovation assumption used in estimation. The mean forecast $\hat{\sigma}_{t,h}$ is obtained as the square root of the average h-step variance, $\hat{\sigma}_{t,h} = \sqrt{h^{-1}\sum_{k=1}^{h}\mathbb{E}_t[\sigma_{t+k}^2]}$. Forecast uncertainty is captured by the in-sample standard deviation s_t of daily volatility over the estimation window. The τ-quantile forecast is then

$$\hat{Q}_{\tau,t}^{\text{GARCH}} = \hat{\sigma}_{t,h} + \Phi^{-1}(\tau)\, s_t, \tag{13}$$

where $\Phi^{-1}(\tau)$ is the standard normal quantile function.

The third benchmark is *HAR-OLS*, the standard HAR model estimated by ordinary least squares. OLS yields only a conditional mean forecast $\hat{\mu}_{t,h}$, but a distributional forecast can be constructed from the in-sample residuals. Let $\hat{\varepsilon}_i = \sigma_i - \hat{\mu}_i$ denote the residuals from the training window. The τ-quantile forecast is then

$$\hat{Q}^{\mathrm{OLS}}_{\tau,t} = \hat{\mu}_{t,h} + \hat{F}_{\varepsilon}^{-1}(\tau), \tag{14}$$

where $\hat{F}_{\varepsilon}^{-1}(\tau)$ is the empirical τ-quantile of the training-window residuals. This pseudo-quantile approach requires no distributional assumption and produces forecasts that are directly comparable to HAR-QR: if HAR-QR provides no improvement over HAR-OLS at quantile τ, then conditioning on the HAR structure suffices and quantile regression adds no value. Comparing the two at $\tau = 0.50$ tests symmetry of the conditional distribution; comparing at $\tau = 0.95$ tests whether the upper tail is better captured by explicit quantile estimation.

3.4 Evaluation Framework

Evaluating volatility forecasts is complicated by the fact that volatility is unobserved, and every evaluation compares forecasts against an imperfect proxy (Patton 2011). We evaluate forecasts against Yang-Zhang volatility computed from realized prices, using loss functions that Patton (2011) has shown to be robust to noise in the volatility proxy.

For mean and median forecasts, we use QLIKE as the primary metric:

$$\mathrm{QLIKE} = \frac{1}{T}\sum_{t=1}^{T}\left(\log \hat{\sigma}_t^2 + \frac{\sigma_t^2}{\hat{\sigma}_t^2}\right). \tag{15}$$

Patton (2011) derives necessary and sufficient conditions for the loss function to yield forecast rankings that are robust to proxy noise, showing that among nine commonly used loss functions, only MSE and QLIKE satisfy these conditions. We prefer QLIKE because it penalizes under-prediction of volatility proportionally more than MSE, which is appropriate when under-prediction is costlier than over-prediction, a property particularly relevant for upper-quantile forecasts. As Puke and Schweikert (2026) demonstrate, aligning estimation and evaluation loss functions, what they call *coherent forecasting*, yields significant accuracy gains. Our HAR-QR models are coherent by construction: the check function (Equation 9) serves as both the estimation objective and the evaluation metric.

As a robustness check, we additionally report mean squared error (MSE) and mean absolute error (MAE) for all point-forecast models:

$$\mathrm{MSE} = \frac{1}{T}\sum_{t=1}^{T}\left(\sigma_t - \hat{\sigma}_t\right)^2, \tag{16}$$

$$\mathrm{MAE} = \frac{1}{T}\sum_{t=1}^{T}\left|\sigma_t - \hat{\sigma}_t\right|. \tag{17}$$

MSE penalizes large errors quadratically and is a member of the proxy-robust class alongside QLIKE (Patton 2011); including it confirms that QLIKE rankings are not sensitive to the choice of loss function within that class. MAE penalizes all deviations linearly and is not proxy-robust, but provides an interpretable measure of average forecast error in the original volatility units.

For quantile forecasts, we use quantile loss:

$$\mathrm{QL}_\tau = \frac{1}{T}\sum_{t=1}^{T} \rho_\tau(\sigma_t - \hat{Q}_{\tau,t}). \tag{18}$$

As a complementary diagnostic, we assess the calibration of quantile forecasts through their *empirical violation rate*. For a given quantile level τ, the violation rate measures the fraction of out-of-sample observations that exceed the quantile forecast:

$$\widehat{VR}_\tau = \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}\left\{\sigma_t > \hat{Q}_{\tau,t}\right\}, \tag{19}$$

where $\mathbf{1}\{\cdot\}$ is the indicator function. A well-calibrated τ-quantile forecast should satisfy $\widehat{VR}_\tau \approx 1 - \tau$. For example, the $\tau = 0.95$ quantile should be exceeded approximately 5% of the time; systematic departures indicate miscalibration. Violation rates above $1-\tau$ signal that the model underestimates the quantile (too narrow), while rates below $1-\tau$ indicate overestimation (too conservative). This diagnostic complements quantile loss, which measures average forecast accuracy but does not directly reveal whether the nominal coverage level is achieved.

To determine whether differences in forecast accuracy are statistically significant, we use the Model Confidence Set (MCS) of Hansen et al. (2011). The MCS identifies the set of models $\mathcal{M}^*$ that contains the best model with a given confidence level, avoiding the need to specify a single benchmark. Starting from the full collection of competing models, the procedure iteratively tests for equal predictive ability and eliminates the worst-performing model until non-rejection occurs. We implement MCS with 10,000 stationary bootstrap replications using the max-t statistic; models in $\mathcal{M}^*$ at the 90% confidence level are considered statistically equivalent.

The forecast evaluation is conducted out-of-sample using a rolling estimation window of 2,520 trading days (approximately 10 years). At each forecast origin t, we estimate parameters using observations from $t - 2519$ to t, generate forecasts, and roll forward by one day. The window rolls *daily* to maximize the number of forecast–realization pairs for statistical evaluation. With approximately 30 years of out-of-sample data, this yields thousands of pairs for statistical testing.

Several assumptions underpin the methodology. First, we assume *local stationarity*: volatility dynamics are stable within each estimation window, though they may shift across windows. The rolling-window design accommodates gradual parameter changes.

Second, we assume that the Yang-Zhang estimator provides a reasonable proxy for latent volatility, which we validate empirically in Section 4. Third, we maintain *linear conditional quantiles* for parsimony, acknowledging that non-linear extensions could potentially improve forecasts. Fourth, we assume ENSO is exogenous to the cocoa market, which is physically reasonable.

Based on the literature, we expect that at short horizons (1 day to 1 month), HAR-QR will match or outperform benchmarks. At long horizons (6–12 months), standard HAR-QR will lose predictive power; we test whether HAR-X-QR augmented with ENSO recovers predictive value, examining whether climate information provides incremental value precisely where the autoregressive signal fades.

Should HAR-X-QR outperform HAR-QR at long horizons, a single full-sample comparison cannot by itself establish whether the gain reflects a structural climate channel or a sample-specific coincidence. We therefore subject any ENSO improvement to three diagnostics, reported in Section 5.5. First, a *temporal-stability* check traces the cumulative ENSO gain as the evaluation horizon advances through time using an expanding-window diagnostic. For each calendar month T in the out-of-sample period, the quantile loss difference $\Delta\mathrm{QL}(T) = \mathrm{QL}_{\text{HAR-X-QR}}(\{t \leq T\}) - \mathrm{QL}_{\text{HAR-QR}}(\{t \leq T\})$ is computed over all forecast–realisation pairs up to and including T. A structural ENSO–volatility channel should produce a steadily accumulating gain throughout the evaluation period, not a gain concentrated in a single episode. Second, a *lag-structure* check re-estimates HAR-X-QR with the Niño 3.4 index lagged 0, 1, 3, and 6 months and inspects the cross-correlogram between ENSO and cocoa volatility. Under the agronomic-transmission hypothesis the predictive content should strengthen at lags of 6–12 months, matching the window over which ENSO-driven rainfall anomalies are hypothesised to affect harvests. This is consistent with the broader finding in the literature that ENSO's predictive value for commodity volatility concentrates at longer horizons: Bouri et al. (2021) document gains at 2–4 year horizons for oil-price realized volatility, while Bonato et al. (2023) find a general tendency for ENSO predictability to strengthen at longer forecast horizons across 16 agricultural commodities. The cross-correlogram follows standard cross-correlation diagnostics for lead–lag relationships between time series (Tsay 2010). Third, a *cross-commodity* check applies the identical HAR-X-QR specification to Arabica coffee and raw sugar, two tropical soft commodities that share weather exposure and downstream markets with cocoa (Ge et al. 2025), since a genuine climate channel should produce comparable gains across commodities grown in the same equatorial belt (Ubilava 2018; Bonato et al. 2023).

4 Data

This section describes the data used to estimate and evaluate the volatility forecasting models developed in Section 3. Two primary datasets are required: daily futures prices for ICE London cocoa contracts and the Niño 3.4 climate index measuring El Niño–Southern Oscillation (ENSO) activity. After describing the cocoa futures data, including contract specifications, the construction of a continuous price series, and the variables derived from it, we turn to the ENSO index used as an exogenous predictor at long horizons. The section concludes with descriptive statistics documenting the stylized facts of cocoa returns and volatility.

4.1 Cocoa Futures Data

The primary data consist of daily Open-High-Low-Close (OHLC) prices and trading volume for ICE Futures Europe cocoa futures contracts. ICE London cocoa futures are standardized contracts for the delivery of 10 metric tonnes of cocoa beans. Each contract specifies delivery in one of five months: March, May, July, September, or December. Prices are quoted in British pounds sterling (GBP) per metric tonne, and trading occurs on the IntercontinentalExchange (ICE) electronic platform during European market hours. The data were obtained from Refinitiv Workspace via the corporate license held by Ritter Sport GmbH & Co. KG.

Futures contracts have finite maturity, requiring the construction of a continuous price series for volatility analysis. We use the generic front-month continuous contract, denoted LCCc1 in the Refinitiv system. This series rolls to the next available contract approximately one month before expiration, maintaining exposure to the most liquid contract while avoiding delivery. The front-month contract typically exhibits the highest trading volume and the tightest bid-ask spreads, making it the standard choice for volatility research on commodity futures (Degiannakis et al. 2022; Alfeus and Nikitopoulos 2022). Rolling between contracts creates artificial price jumps when the new contract trades at a different level than the expiring one (the roll yield). For volatility estimation using OHLC data, these jumps are largely captured within the daily range and do not require explicit adjustment. The Yang-Zhang estimator (Section 3.1) is specifically designed to be robust to opening jumps, which include both overnight news and contract roll effects.

The daily dataset spans 10,376 trading days from January 2, 1985, to December 15, 2025. This extended sample offers several advantages. It encompasses multiple El Niño and La Niña cycles, providing sufficient variation in the ENSO predictor for long-horizon models. It includes diverse market regimes: the long period of relative stability from 1990–2010, the financialization era beginning around 2008, and the recent 2022–2024 price surge attributed to the Cocoa Swollen Shoot Virus Disease (CSSVD) outbreak and other supply-side disruptions in West Africa (Institute 2024; Organization 2024). The sample also includes approximately one year of high-frequency data (February 3, 2025, through January 12, 2026, comprising 20,791 five-minute price observations), which enables validation of the Yang-Zhang volatility estimator against realized volatility computed from intraday returns.

Table 12 in Appendix A summarizes the coverage and key characteristics of the cocoa futures data.

For each trading day t, the dataset provides five price observations: the opening price O_t, the highest price during the trading session H_t, the lowest price L_t and the closing price C_t. Log returns $r_t = \ln(C_t/C_{t-1})$ are computed for descriptive statistics and as inputs to the GARCH benchmark.

From the daily OHLC prices, we construct the Yang-Zhang volatility series (Equation (1)), which serves as the dependent variable in all forecasting models and as the realized measure against which forecasts are evaluated.

For the one-year period with available intraday data, we compute realized volatility as the sum of squared five-minute log returns:

$$\mathrm{RV}_t = \sum_{i=1}^{M} r_{t,i}^2, \tag{20}$$

where $r_{t,i}$ is the log return for the i-th five-minute interval on day t, and M is the number of five-minute intervals in the trading day (approximately 78 intervals for a 6.5-hour trading session). This realized volatility measure serves as the benchmark in a Mincer-Zarnowitz validation of the Yang-Zhang estimator over the recent subsample.

On data quality, the daily series contain no missing values for trading days; non-trading days are excluded. Overnight returns spanning weekends or holidays are not rescaled by calendar days elapsed, as the larger return reflects genuine accumulated risk rather than a measurement artefact.

4.2 ENSO Climate Data

As discussed in Section 2.5, ENSO is hypothesised to provide a physically motivated predictor for cocoa volatility at long horizons. We measure ENSO activity using the Niño 3.4 index, a measure of sea surface temperature anomalies in the central equatorial Pacific (5°N–5°S, 170°W–120°W), computed as a three-month running mean relative to a climatological base period. Monthly values are obtained from the NOAA Climate Prediction Center, calculated using the ERSSTv6 dataset and publicly available at `https://psl.noaa.gov/data/correlation/nina34.data`.

Our sample covers January 1985 through December 2025, providing 492 monthly observations. NOAA classifies an El Niño (La Niña) episode when the three-month running mean of Niño 3.4 anomalies exceeds +0.5 °C (falls below −0.5 °C) for at least five consecutive overlapping three-month periods (Lindsey 2009). In the regression models (Equation 7), ENSO_t enters as the continuous Niño 3.4 value rather than a categorical regime indicator, preserving the full variation in anomaly magnitude.

The sample contains 11 El Niño and 12 La Niña episodes over the 1985–2025 period. Figure 1 plots the Niño 3.4 index over the cocoa sample period; red shading marks El Niño episodes (index $> +0.5$ °C) and blue shading marks La Niña episodes (index < -0.5 °C), with the horizontal dashed lines indicating the NOAA classification thresholds.

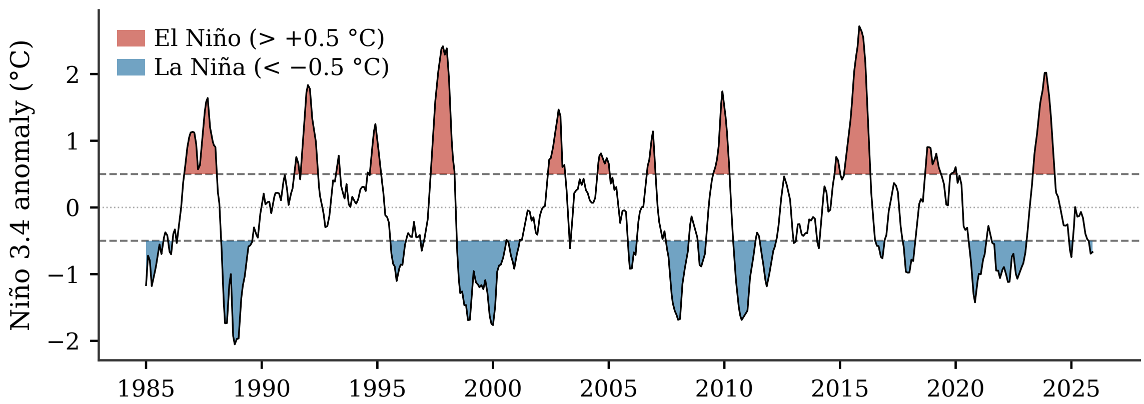

Figure 1: Niño 3.4 Sea Surface Temperature Anomaly (1985–2025)

Monthly Niño 3.4 index (three-month running mean, ERSSTv6 base period). Red: El Niño episodes (anomaly $> +0.5$ °C for at least five consecutive months); blue: La Niña (anomaly < -0.5 °C). Dashed lines: ± 0.5 °C thresholds. Source: NOAA Climate Prediction Center.

Since the index is monthly while cocoa futures trade daily, each trading day inherits the Niño 3.4 value of its calendar month, as described in Section 3.2. The data contain no missing values through December 2025, covering the full cocoa futures sample period.

4.3 Descriptive Statistics and Stylized Facts

This subsection presents the empirical characteristics of cocoa returns and volatility, establishing the stylized facts that motivate the modeling choices in Section 3.

Figure 2 displays the daily closing price of the front-month cocoa futures contract over the full sample period. The series exhibits several distinct episodes. From the mid-1980s through the early 2000s, prices declined from their earlier highs and stabilized, fluctuating around a mean of approximately 1,300 GBP per tonne through the 1990s and 2000s. The 2010s showed gradual price increases, followed by the recent 2022–2024 surge reaching over 10,000 GBP per tonne, unprecedented in the modern trading history of cocoa, driven by the CSSVD outbreak in key producing regions.

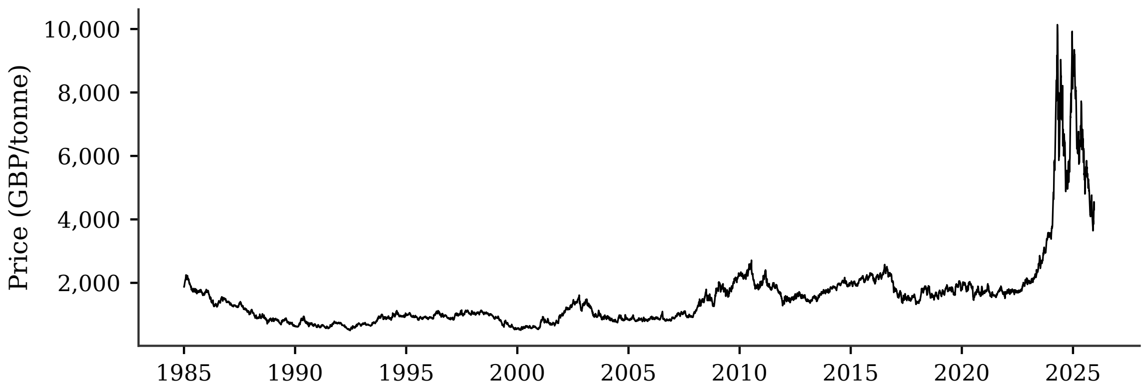

Figure 2: ICE London Cocoa Futures Daily Closing Price (1985–2025)

Front-month continuous contract (LCCc1), daily closing prices in GBP per metric tonne. Source: Refinitiv Workspace. Sample: 2 January 1985 – 15 December 2025 (10,376 trading days).

Table 2 presents descriptive statistics for daily log-returns. The mean daily return is slightly positive (0.008%), corresponding to an approximate annualized mean return of 2% over the full sample. However, returns exhibit substantial volatility: the standard deviation of 1.82% implies an annualized volatility of approximately 29%, consistent with cocoa's classification as a high-volatility commodity. The distribution of returns deviates sharply from normality. Negative skewness (-0.37) indicates a left tail longer than the right, reflecting the tendency for large negative price movements to occur more frequently than large positive ones. Excess kurtosis of 9.06, formally confirmed by the Jarque-Bera statistic of 35,652 ($p < 0.001$), documents fat tails: extreme returns occur far more often than a Gaussian distribution predicts, implying that volatility models assuming Gaussian innovations may understate tail risk.

Turning to the volatility series, Figure 3 displays the Yang-Zhang volatility estimator (annualized, in percent) computed from daily OHLC data. The figure reveals pronounced volatility clustering: periods of high volatility persist for extended intervals before reverting to lower levels. The 2022–2024 supply crisis stands out as the dominant volatility regime in the sample, with annualized volatility exceeding 100% at its peak. Between this episode and the mid-1980s, volatility fluctuates around a lower baseline with occasional spikes corresponding to supply shocks, weather events, and financial market turbulence (e.g., the 2008 financial crisis).

The persistence of volatility justifies the HAR model's multi-horizon structure: current volatility depends not only on yesterday's volatility (daily component) but also on average volatility over the past week and month (weekly and monthly components). The clustering pattern also motivates quantile regression: during high-volatility regimes, the tails of the volatility distribution shift outward, and point forecasts of the mean may underestimate the risk of extreme outcomes.

The regression models in Section 3 require stationary inputs. Augmented Dickey–

Table 2: Descriptive Statistics for Daily Log-Returns

Statistic	**Value**
Observations	10,375
Mean (%)	0.008
Median (%)	0.000
Std. Deviation (%)	1.816
Minimum (%)	−25.385
Maximum (%)	11.585
25th Percentile (%)	−0.903
75th Percentile (%)	0.915
Skewness	−0.368
Excess Kurtosis	9.057
Jarque-Bera	35,652
p-value	< 0.001

Daily log-returns $r_t = \ln(C_t/C_{t-1})$, ICE London cocoa front-month contract (LCCc1), January 1985–December 2025. Values in percentage points. Excess kurtosis reported (normal distribution: 0). Jarque-Bera tests joint null of zero skewness and zero excess kurtosis.

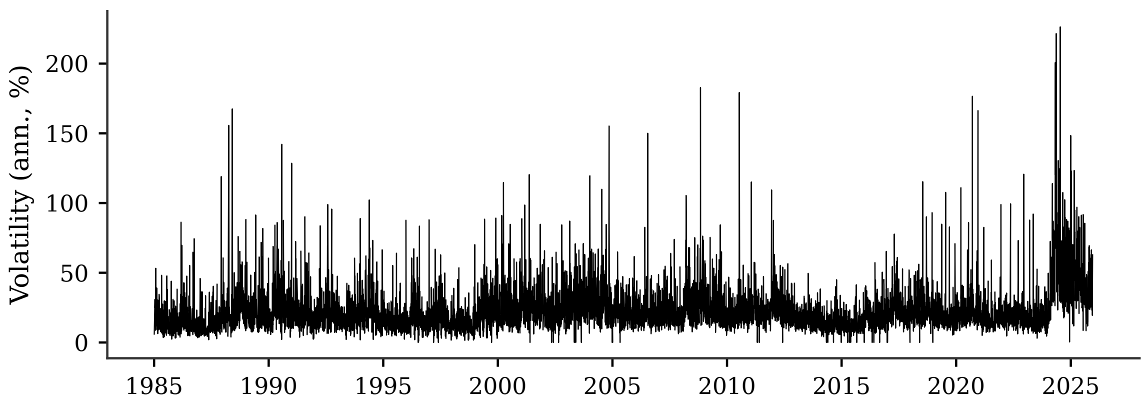

Figure 3: Yang-Zhang Annualized Volatility Time Series (1985–2025)

Yang-Zhang volatility (Yang and Zhang 2000) computed from daily OHLC prices, annualized (percent). Combines overnight, intraday, and close-to-close variance components.

Fuller and Phillips–Perron tests reject the unit root null at the 1% level for all three series (Table 13, Appendix A), justifying the use of standard OLS and quantile regression without differencing or cointegration corrections.

The data described in this section provide the foundation for the empirical analysis. The extended daily sample (1985–2025) ensures sufficient variation in both volatility regimes and ENSO phases, while the intraday subsample enables proxy validation. A Mincer-Zarnowitz regression of intraday realized volatility on the Yang-Zhang estimator (Table 14, Appendix A) yields $R^2 = 0.847$ and $\hat{\beta} = 0.836$ at the 22-day frequency relevant to the HAR monthly component, confirming strong co-movement; the lower $\hat{\beta}$ at the daily frequency reflects the overnight variance component that intraday RV does not capture. The next section presents the empirical results.

5 Empirical Results

This section presents the out-of-sample forecasting results, organized around the two research questions. Section 5.1 compares mean forecast accuracy across the three baseline models. Section 5.2 evaluates distributional forecasts at the upper tail, testing whether HAR-QR outperforms the benchmarks across horizons (RQ1). Section 5.3 assesses whether the Niño 3.4 climate index adds forecast value at long horizons (RQ2). Section 5.4 tests the statistical significance of the observed differences using Model Confidence Sets. Section 5.6 synthesizes the findings. All results use a rolling estimation window of 2,520 trading days with daily re-estimation, yielding approximately 7,800 out-of-sample forecast–realization pairs at each horizon.

5.1 Mean Forecast Comparison

Table 3 reports the QLIKE loss for the three mean forecast models across all six horizons. The ranking is unambiguous: HAR-OLS achieves the lowest (best) QLIKE at every horizon, followed by GARCH(1,1), with historical volatility uniformly last.

Table 3: Mean forecast QLIKE loss by model and horizon

Model	1d	1w	1m	3m	6m	12m
Historical Vol	−7.156	−7.367	−7.420*	−7.405	−7.380	−7.353
GARCH(1,1)	−7.239	−7.379	−7.401*	−7.375	−7.348	−7.336
HAR-OLS	**−7.291**	**−7.460**	**−7.493***	**−7.474**	**−7.438**	**−7.423**

QLIKE loss (Patton 2011); more negative indicates better performance. Bold: best model at each horizon. *Best horizon for each model.

The gap between HAR-OLS and GARCH is largest at the one-month horizon (0.092 QLIKE units) and narrowest at one day (0.052 units). The monthly HAR component ($\sigma_t^{(m)}$) maintains a correlation of 0.66 with the twelve-month volatility target, compared to 0.37 for the daily component, an empirical fact that points to the source of HAR-OLS's long-horizon advantage.

The relative performance of the two benchmarks reverses across horizons: at one day, GARCH outperforms historical volatility (−7.239 vs. −7.156), but from one month onward historical volatility is the stronger benchmark, eventually surpassing GARCH by 0.017 QLIKE units at twelve months. All three models share a common horizon pattern: loss improves from the one-day to the one-month horizon, reflecting the smoothing effect of averaging in the target $\bar{\sigma}_{t:t+h}$, before degrading at six and twelve months as the autoregressive signal weakens. Even HAR-OLS degrades beyond one month, which motivates the search for additional predictive signals in Section 5.3. The mechanism behind the GARCH and HAR ranking is interpreted in Section 6.1.

The QLIKE ranking is confirmed by MSE and MAE. Figure 4 plots all three loss functions across horizons. HAR-OLS achieves the lowest MSE at every horizon; MAE is minimized by HAR-X-QR (median) at short horizons and by HAR-OLS or HAR-QR (median) beyond three months. Crucially, GARCH(1,1) is the worst-performing model under all three metrics at long horizons, with its error rising steeply relative to the HAR family as the horizon extends, a direct consequence of mean-reversion in its multi-step forecasts. The HAR family dominates regardless of how point forecast errors are penalized.

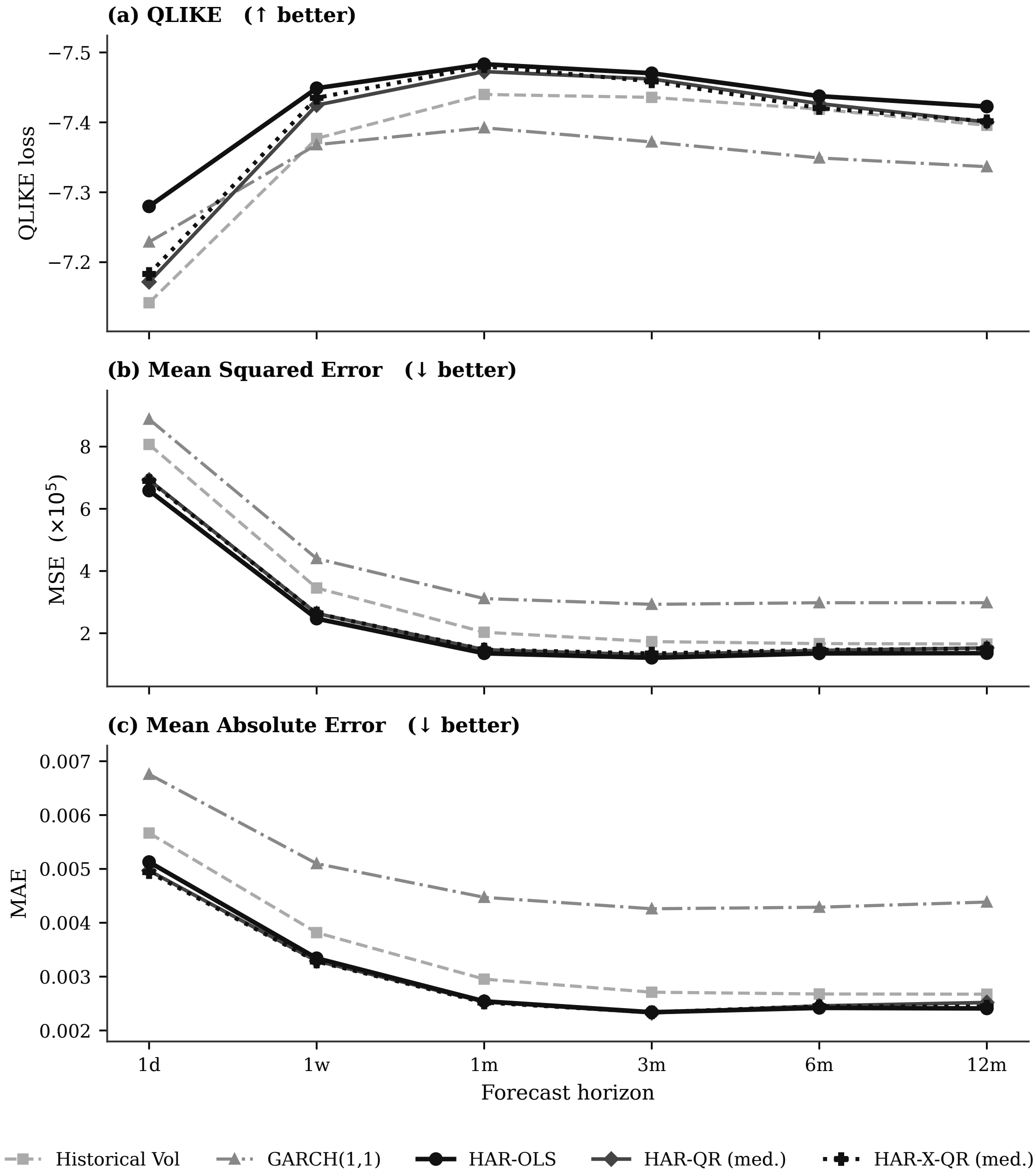

Figure 4: Robustness check: point forecast accuracy under three loss functions
Each panel plots mean (or median) forecast loss across horizons. QLIKE axis is inverted (more negative = better). MSE scaled by 10^5 for readability. The consistent separation between GARCH(1,1) and the HAR family at long horizons holds across all three metrics.

5.2 Quantile Forecast Comparison (RQ1)

Mean forecasts summarize only the center of the conditional distribution. For risk assessment, the upper tail matters: a model that accurately forecasts average volatility may still understate the probability of extreme outcomes. We now compare five models on their ability to forecast the 95th percentile of future volatility. Historical volatility uses the empirical quantile of the trailing 252-day distribution. GARCH(1,1) quantiles follow the closed-form Gaussian approach (Equation 13). HAR-OLS produces pseudo-quantile forecasts from its in-sample residual distribution (Equation 14). HAR-QR estimates the quantile directly through quantile regression on the HAR structure, and HAR-X-QR augments this with the Niño 3.4 index.

Table 4 reports the quantile loss at $\tau = 0.95$ for all five models.

Table 4: Upper-tail quantile loss ($\tau = 0.95$) by model and horizon

Model	1d	1w	1m	3m	6m	12m
Historical Vol	0.001282	0.000880	0.000769*	0.000803	0.000843	0.000862
GARCH(1,1)	0.001307	0.000972	0.000905	0.000956	0.000971	0.000932*
HAR-OLS	0.001228	0.000733	0.000509	0.000493*	0.000513	0.000493*
HAR-QR	**0.001160**	**0.000680**	**0.000470**	0.000461*	0.000510	0.000516
HAR-X-QR	0.001161	0.000681	0.000476	**0.000442**	**0.000419**	**0.000357***

Quantile loss (Equation (18)). Lower values indicate better forecasts. Bold: best model at each horizon. *Best horizon for each model.

Four patterns emerge from this comparison.

First, the HAR family dominates all benchmarks at every horizon. HAR-QR, HAR-OLS, and HAR-X-QR all outperform both GARCH(1,1) and historical volatility at every horizon. The gap is not marginal: at twelve months, the best benchmark (historical volatility, 0.000862) is 67% above HAR-QR (0.000516) and 142% above HAR-X-QR (0.000357).

Second, direct quantile estimation outperforms the pseudo-quantile approach at short-to-medium horizons. HAR-QR improves on HAR-OLS by 6% at one day (0.001160 vs 0.001228) and 8% at one month (0.000470 vs 0.000509). The advantage narrows toward six months and reverses marginally at twelve months, where HAR-OLS holds a slight numerical edge (0.000493 vs 0.000516). Whether this reversal is statistically meaningful is assessed in Section 5.4.

Third, the GARCH quantile forecast trails the HAR family from one week onward. At one day, GARCH (0.001307) and historical volatility (0.001282) are similar, with historical volatility slightly better. From one week onward, GARCH produces the highest quantile loss of all five models; by twelve months, its quantile loss (0.000932) is 80% above HAR-QR's (0.000516).

Fourth, HAR-X-QR separates from HAR-QR only at long horizons. At one day through one month, the two models are virtually indistinguishable (differences < 1%). The divergence begins at three months (+4.17% improvement) and accelerates to +17.90% at six months and +30.86% at twelve months. This horizon dependence is analyzed in detail in Section 5.3.

Figure 5 illustrates the distributional forecasts of the HAR-X-QR model, displaying the 50–75% and 75–95% quantile bands alongside the median forecast ($\tau = 0.50$) and Yang-Zhang volatility over the full out-of-sample period (1995–2025). The figure complements the loss-function evidence by showing how the model's forecast distribution moves through regimes.

At the one-month horizon (panel a), the quantile bands widen and narrow in step with Yang-Zhang volatility, demonstrating the HAR structure's ability to adapt to changing variance regimes. At the twelve-month horizon (panel b), the bands are smoother, reflecting the averaging inherent in long-run projection, but still widen noticeably ahead of the 2022–2024 supply crisis. The Yang-Zhang series breaches the $\tau = 0.95$ boundary during the crisis peak at both horizons, consistent with the violation rates reported in Table 5.

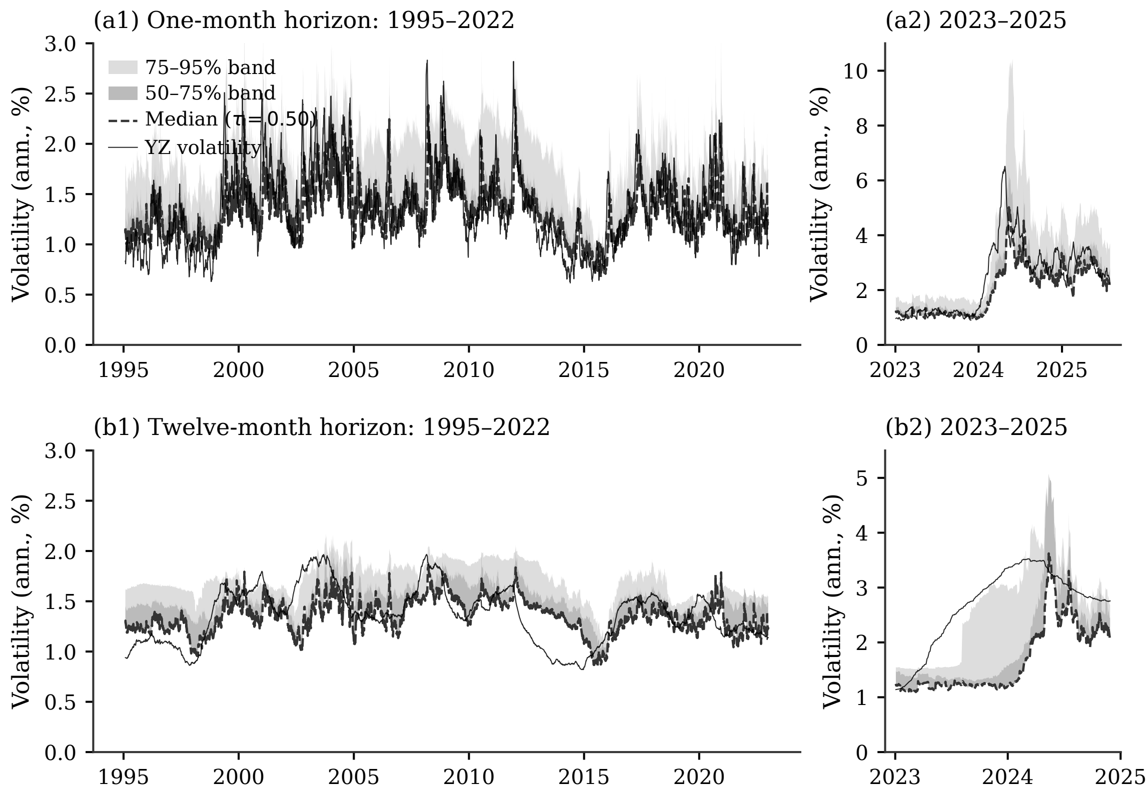

Figure 5: HAR-X-QR distributional forecasts: quantile bands and Yang-Zhang volatility

HAR-X-QR quantile forecasts at $\tau \in \{0.50, 0.75, 0.95\}$ plotted against Yang-Zhang volatility (solid black). Dark gray band: 50–75% interval; light gray band: 75–95% interval; dashed line: median ($\tau = 0.50$). Rows: one-month (a) and twelve-month (b) horizons.

Table 5 reports violation rates for the upper quantile ($\tau = 0.95$) by horizon for both HAR-QR and HAR-X-QR. Under correct unconditional coverage, the violation rate should equal $1 - \tau = 5\%$. HAR-QR achieves near-target coverage at one day and one week (5.17% and 5.30%), remains acceptable through three months (7.05%), and exceeds the target at six months (8.57%) and twelve months (10.36%). ENSO augmentation leaves short-horizon coverage essentially unchanged but widens the long-horizon gap: HAR-X-QR violation rates rise to 10.97% at six months and 12.66% at twelve months. The apparent tension between HAR-X-QR's lower quantile loss and its higher violation rate, and the mechanism that produces it, is taken up in Section 6.4. The following subsection examines whether augmenting HAR-QR with the ENSO climate index improves long-horizon upper-tail accuracy.

Table 5: Upper-quantile ($\tau = 0.95$) coverage by horizon: HAR-QR and HAR-X-QR

Horizon	N	HAR-QR rate (%)	HAR-X-QR rate (%)
1d	7,833	5.17	5.14
1w	7,829	5.30	5.35
1m	7,812	6.22	6.90
3m	7,768	7.05	7.90
6m	7,702	8.57	10.97
12m	7,570	10.36	12.66

A violation occurs when Yang-Zhang volatility exceeds the $\tau = 0.95$ quantile forecast; the nominal target is 5% ($= 1 - \tau$). N: forecast–realization pairs per horizon.

5.3 ENSO Augmentation: HAR-X-QR (RQ2)

Table 6 reports the percentage change in quantile loss when moving from HAR-QR to HAR-X-QR across all three quantile levels and six horizons. Positive values indicate improvement (lower loss under HAR-X-QR).

Table 6: Quantile loss improvement (%) from ENSO augmentation (HAR-X-QR vs. HAR-QR)

τ	1d	1w	1m	3m	6m	12m
0.50	+0.5713	+0.2694	+0.4893	−0.4268	+0.2707	+3.0358
0.75	+0.4751	+0.0543	−0.1509	−1.4625	−0.6454	−1.1074
0.95	+0.0185	−0.0277	−1.0514	+4.1698	+17.9025	+30.8611

Improvement (%) $= (1 - \mathrm{QL}_{\text{HAR-X}}/\mathrm{QL}_{\text{HAR}}) \times 100$; positive = lower loss for HAR-X-QR.

The results reveal a striking horizon- and quantile-dependent pattern. At short horizons (one day to one month), ENSO augmentation provides no benefit and occasionally worsens forecasts marginally. At such horizons, the Niño 3.4 coefficient captures noise rather than signal, and the additional parameter slightly degrades out-of-sample performance.

The first meaningful improvement appears at three months for the upper tail: $\tau = 0.95$ quantile loss decreases by 4.17%. The effect strengthens substantially at six months (+17.90%) and reaches its maximum at twelve months (+30.86%). This concentration in the upper tail is consistent with the hypothesis that ENSO anomalies are associated with cocoa supply disruptions, which manifest as rightward shifts in the volatility distribution rather than a uniform upward shift. Whether this recalibration reflects a stable predictive relationship or a sample-specific regime effect is assessed in Section 5.5. The GARCH and historical volatility benchmarks cannot exploit this climate index, explaining why their twelve-month upper-tail performance in Table 4 falls far behind HAR-X-QR: historical volatility's loss (0.000862) is 142% higher than HAR-X-QR's (0.000357), and GARCH's (0.000932) is 161% higher.

The median ($\tau = 0.50$) shows only a small gain at twelve months (+3.04%), with negligible or marginally negative effects at shorter horizons, confirming that the benefit of ENSO augmentation is predominantly distributional rather than centered on the mean.

5.4 Statistical Significance: Model Confidence Sets

The preceding sections compare models by point estimates of loss functions. To assess whether the observed differences are statistically significant, we apply the Model Confidence Set (MCS) procedure of Hansen et al. (2011). The MCS identifies the set $\mathcal{M}^*$ of models whose forecasting ability cannot be distinguished from that of the best model at a given confidence level. We use the MCS procedure and configuration described in Section 3.4.

Table 7 reports MCS results for QLIKE loss. HAR-OLS is the sole member of $\mathcal{M}^*$ at the one-day and one-week horizons ($p = 1.00$), confirming that its mean forecast superiority is statistically significant at short horizons. From one month onward, historical volatility, HAR-QR, and HAR-X-QR enter $\mathcal{M}^*$ alongside HAR-OLS, indicating that at medium and long horizons their mean forecast accuracy is statistically indistinguishable from HAR-OLS. GARCH(1,1) is excluded from $\mathcal{M}^*$ at horizons up to six months; at twelve months it enters $\mathcal{M}^*$ as well, reflecting the convergence of all models in the tails of the QLIKE distribution at the longest horizon.

Table 8 presents MCS results for quantile loss at $\tau = 0.95$, the central evaluation criterion for distributional forecasting. Historical volatility and GARCH(1,1) are excluded from $\mathcal{M}^*$ at every horizon, confirming that their inferior quantile loss is statistically significant and not a product of sampling variation. From one month onward, $\mathcal{M}^*$ contains HAR-OLS alongside the two direct quantile regression models, indicating

Table 7: Model Confidence Set: QLIKE loss (mean forecasts)

Model		Horizon 1d	1w	1m	3m	6m	12m
Historical Vol	p	0.05	0.08	0.12	0.18	0.19	0.22
	$\mathcal{M}^*$			✓	✓	✓	✓
GARCH(1,1)	p	0.05	0.03	0.01	0.01	0.07	0.22
	$\mathcal{M}^*$						✓
HAR-OLS	p	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
	$\mathcal{M}^*$	✓	✓	✓	✓	✓	✓
HAR-QR (med.)	p	0.05	0.08	0.12	0.20	0.23	0.22
	$\mathcal{M}^*$			✓	✓	✓	✓
HAR-X-QR (med.)	p	0.05	0.08	0.12	0.20	0.23	0.22
	$\mathcal{M}^*$			✓	✓	✓	✓

MCS p-values (Hansen et al. 2011): max-t statistic, 10,000 bootstrap replications, 90% confidence. ✓: $\mathcal{M}^*$ membership. Bold: $p = 1.00$ (best-ranked model). HAR-QR and HAR-X-QR enter as median ($\tau = 0.50$) forecasts.

that the pseudo-quantile approach is statistically indistinguishable from HAR-QR at medium and long horizons. At short horizons, HAR-OLS is excluded ($p = 0.06$ at one day, $p = 0.07$ at one week), confirming that explicit quantile estimation provides a statistically significant advantage where volatility dynamics are most state-dependent.

Within the HAR family, a revealing pattern emerges. At the one-day horizon, both HAR-QR ($p = 1.00$) and HAR-X-QR ($p = 0.36$) are in $\mathcal{M}^*$, with HAR-QR ranked first. From one week onward, both models coexist in $\mathcal{M}^*$. The critical shift occurs at three months: HAR-X-QR takes $p = 1.00$ from HAR-QR, indicating that the ENSO-augmented model becomes the top-ranked specification. This reversal persists at six and twelve months ($p = 1.00$ for HAR-X-QR versus $p = 0.11$ and $p = 0.15$ for HAR-QR). Both models remain in $\mathcal{M}^*$ throughout, meaning the data do not permit a statistically significant ranking between them. The consistent shift of the top rank to HAR-X-QR from three months onward is descriptively suggestive, but any interpretation as evidence for a structural ENSO channel requires a robustness assessment.

Table 8: Model Confidence Set: quantile loss at $\tau = 0.95$

Model		Horizon					
		1d	1w	1m	3m	6m	12m
Historical Vol	p	0.06	0.00	0.00	0.00	0.00	0.00
	$\mathcal{M}^*$						
GARCH(1,1)	p	0.00	0.00	0.00	0.00	0.00	0.00
	$\mathcal{M}^*$						
HAR-OLS	p	0.06	0.07	0.47	0.51	0.30	0.21
	$\mathcal{M}^*$			✓	✓	✓	✓
HAR-QR	p	**1.00**	**1.00**	**1.00**	0.51	0.11	0.15
	$\mathcal{M}^*$	✓	✓	✓	✓	✓	✓
HAR-X-QR	p	0.36	0.82	0.52	**1.00**	**1.00**	**1.00**
	$\mathcal{M}^*$	✓	✓	✓	✓	✓	✓

Same MCS configuration as Table 7. ✓: $\mathcal{M}^*$ membership at 90% confidence. Bold: $p = 1.00$ (best-ranked model).

5.5 Robustness of the ENSO Signal

The preceding results establish that HAR-X-QR outperforms HAR-QR over the full evaluation sample, particularly at the twelve-month upper tail. Before interpreting this as evidence of a structural ENSO–cocoa volatility channel, we apply the three robustness diagnostics outlined in Section 3: temporal stability, lag structure, and cross-commodity replication. Each addresses a distinct way in which the full-sample gain could be sample-specific rather than generalisable.

Temporal stability: expanding-window diagnostic

To assess whether the ENSO gain is concentrated in the 2022–2024 supply crisis, we trace the *cumulative* gain as the evaluation horizon advances through time. For each calendar month T in the out-of-sample period, define

$$\Delta\mathrm{QL}(T) \;=\; \mathrm{QL}_{\text{HAR-X-QR}}\Big(\{t \le T\}\Big) \;-\; \mathrm{QL}_{\text{HAR-QR}}\Big(\{t \le T\}\Big),$$

where each loss is computed over all forecast–realisation pairs with evaluation date up to and including T. Negative values indicate that ENSO augmentation has been beneficial on a cumulative basis up to that point. The construction is fully recursive: no break date is assumed or imposed, and the curve spans the complete out-of-sample window without any ex-post partitioning.

Figure 6 shows that the full-sample ENSO gain is almost entirely a product of the 2022–2024 extreme-volatility episode, and that this concentration is specific to the upper tail.

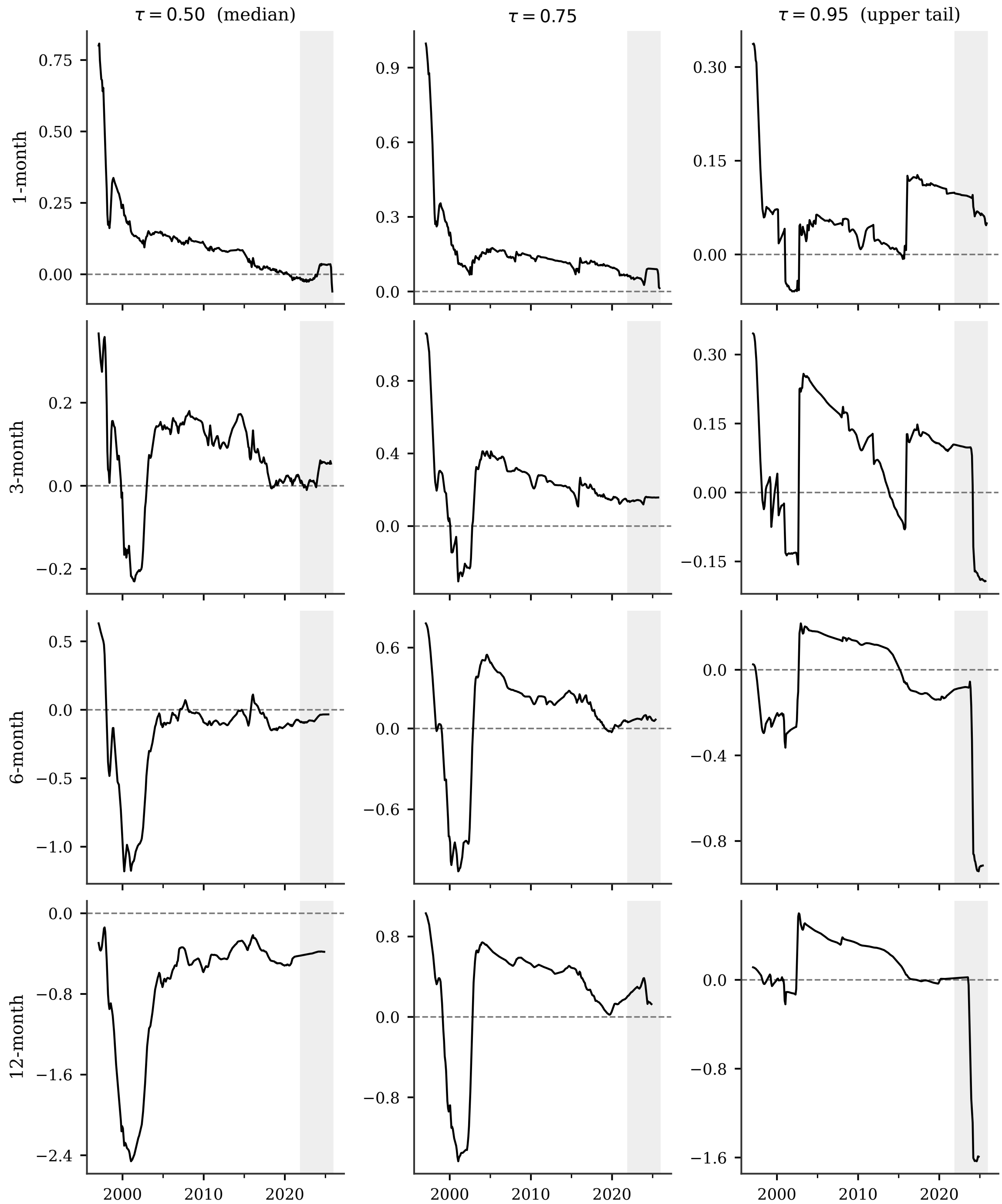

Figure 6: Expanding-window cumulative ENSO augmentation gain (ΔQL) across horizons and quantile levels

Each panel shows $\Delta\mathrm{QL}(T)$ over all out-of-sample pairs with evaluation date $\leq T$; negative values indicate cumulative net gain from ENSO augmentation. Rows: horizons (1, 3, 6, 12 months). Columns: $\tau \in \{0.50, 0.75, 0.95\}$. Shaded region: post-2022 supply-crisis episode (no formal cutoff assumed). Minimum 500 observations before plotting.

At the twelve-month horizon and $\tau = 0.95$, the cumulative ΔQL remains positive (ENSO augmentation harmful on net) or indistinguishable from zero throughout the period prior to 2022. The curve then falls sharply during 2023 and 2024, with the entire full-sample improvement of 30.8% accumulating within those two years. At the six-month horizon the pattern is similar: a modest negative signal builds gradually from around 2014, but the dominant movement occurs in the 2023–2024 window, when the six-month curve falls from near zero to its full-sample endpoint of $\Delta\text{QL} = -0.91 \times 10^{-4}$.

The median ($\tau = 0.50$) and $\tau = 0.75$ columns tell a qualitatively different story. Across all four horizons those panels show irregular, oscillating paths that cross zero repeatedly and never accumulate a sustained downward trend, even during the 2022–2024 crisis window. The ENSO augmentation gain is therefore not a broad distributional shift but a recalibration concentrated in the extreme upper tail. This is consistent with the mechanism described in Section 5.3: the positive ENSO coefficient raises the $\tau = 0.95$ boundary during El Niño episodes, reducing the magnitude of large upside exceedances, while leaving the center of the distribution largely unaffected.

This result sharpens the robustness assessment. The expanding-window diagnostic shows *when* the gain first accumulates and *where* in the distribution it is concentrated. For the twelve-month upper tail, the answer to both questions is the same: almost exclusively during the 2022–2024 supply crisis. The reported 30.8% full-sample improvement reflects a gain that accrued in the final two years of the evaluation period and is absent from the median and $\tau = 0.75$ forecasts throughout.

Lag structure

If ENSO drives cocoa volatility through the agronomic-transmission chain (ENSO anomaly $\rightarrow$ West African rainfall $\rightarrow$ growing-season disruption $\rightarrow$ harvest shortfall $\rightarrow$ supply tightening $\rightarrow$ price volatility), the predictive power of the Niño 3.4 index should *strengthen* as the lag approaches 6–12 months, the time over which this chain unfolds. To test this, Table 9 reports quantile loss at $\tau = 0.95$ for HAR-X-QR with ENSO lagged 0, 1, 3, and 6 months.

Table 9: Lag-decay diagnostic: quantile loss ($\tau = 0.95$) by ENSO lag

	ENSO lag (months)			
Horizon	0	1	3	6
1m	0.000476	0.000476	**0.000467**	0.000470
6m	0.000419	0.000409	**0.000408**	0.000443
12m	0.000357	**0.000343**	0.000384	0.000466
HAR-QR baseline	0.000470	0.000510	0.000516	—

Lower values indicate better forecasts. Bold: best lag for each horizon. Lag k: Niño 3.4 from k calendar months prior to the forecast date; construction is fully causal.

The results contradict the physical-channel prediction. At the twelve-month horizon, predictive power *peaks at lag 1* (0.000343) and decays sharply at lags 3 (0.000384) and 6 (0.000466), the latter substantially above the lag-1 value. At the six-month horizon, the pattern is similar: the best lag is 3 (0.000408), and lag 6 already deteriorates to 0.000443, well above the contemporaneous value. Under the agronomic-transmission hypothesis, lags 6 and 12 should be the most informative; the data show the opposite. The contemporaneous and near-contemporaneous specification outperforms all longer lags. The interpretation of this lag structure is taken up in Section 6.2.

This is further confirmed by the cross-correlogram in Figure 7, which plots the Pearson correlation between monthly Niño 3.4 and monthly mean cocoa volatility at lags −24 to +24 months. Positive lags indicate that ENSO leads volatility. In the full sample, the correlation is statistically significant only at lags 0, 1, and 2 on the positive side; the physical-channel prediction window (lags +6 to +12) is entirely insignificant. In the subsample excluding the supply-crisis period, correlations are significant across a wider range of lags, but this reflects the strong autocorrelation of the Niño 3.4 index itself: once ENSO is negative, it tends to remain so for 6–12 months, inducing apparent significance at distant lags through ENSO's own persistence rather than through independent predictive relationships.

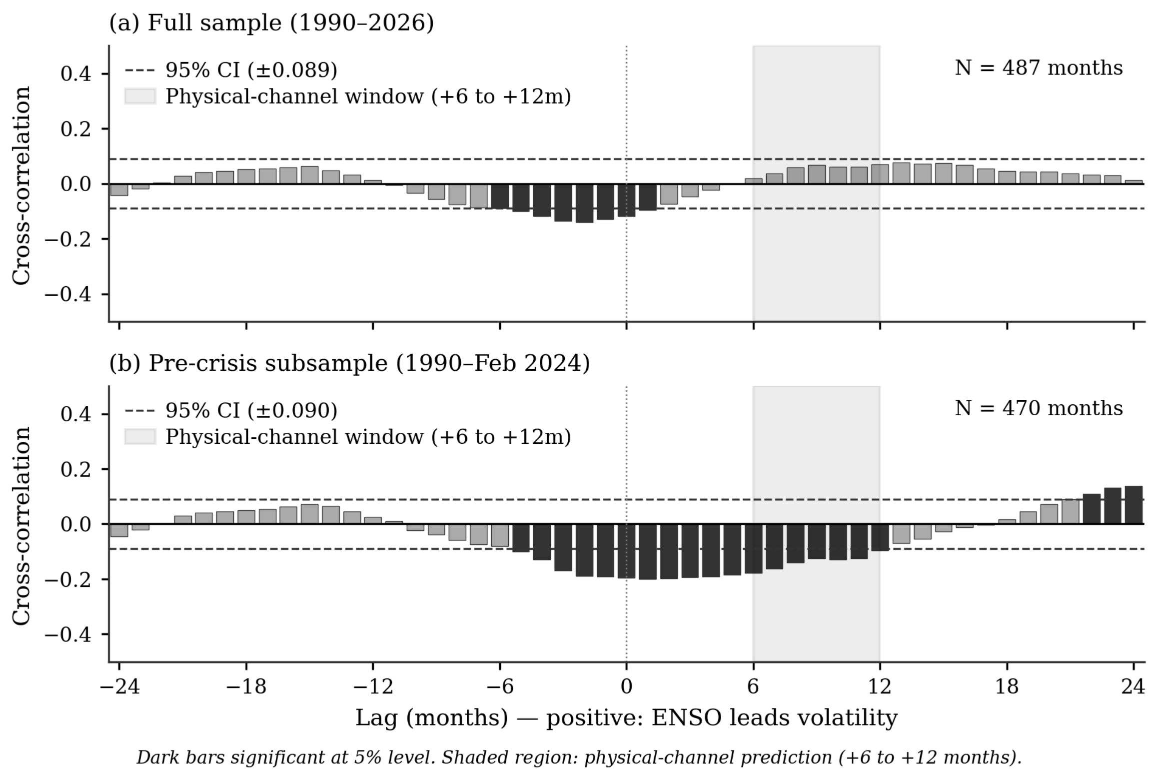

Figure 7: Cross-correlogram: Niño 3.4 vs. monthly mean cocoa OHLC volatility
Pearson correlation at lags −24 to +24 months. Positive lag k: Niño 3.4 at time t leads volatility at $t+k$. Dark bars exceed the 95% confidence band ($\pm 1.96/\sqrt{N}$). Shaded region: physical-channel window (+6 to +12 months). Panel (a): full sample; panel (b): subsample excluding 2022 onward.

Cross-commodity validation

If ENSO operates as a genuine climate channel it should produce comparable predictive gains on other tropical commodities exposed to the same Pacific climate signal. Arabica coffee and raw sugar are natural comparators: both are weather-sensitive soft commodities that share downstream markets and exhibit volatility spillovers with cocoa (Ge et al. 2025). Table 10 reports the $\tau = 0.95$ quantile loss improvement from ENSO augmentation for Arabica coffee (ICE KC=F, USD/lb) and raw sugar (ICE SB=F, USD cents/lb), estimated using the identical HAR-X-QR specification and rolling-window setup.

Table 10: Cross-commodity ENSO gain (ΔQL, $\tau = 0.95$) at 6- and 12-month horizons

Horizon	Cocoa	Coffee	Sugar
6m	**−0.000091**	+0.000014	−0.000014
12m	**−0.000159**	+0.000031	−0.000005

$\Delta\text{QL} = \text{QL}_{\text{HAR-X-QR}} - \text{QL}_{\text{HAR-QR}}$; negative values indicate improvement from ENSO augmentation. Coffee: ICE KC=F, $N \approx 3{,}870$ pairs. Sugar: ICE SB=F, $N \approx 3{,}832$ pairs.

ENSO augmentation does not replicate on either comparator. For coffee, ENSO *worsens* long-horizon upper-tail forecasts at both horizons, with a +0.000031 deterioration at twelve months. For sugar, the gain is −0.000005 at twelve months, a factor of 32 smaller than cocoa's −0.000159.

Summary assessment of RQ2 robustness

The three diagnostics yield consistent empirical findings. The cumulative ENSO gain at the twelve-month upper tail is indistinguishable from zero throughout the three decades preceding the 2022–2024 supply crisis, and the entire 30.8% full-sample improvement accumulates within the 2023–2024 window. The expanding-window matrix shows that the gain is present only at $\tau = 0.95$ and at horizons of six months and beyond; the median and $\tau = 0.75$ forecasts show no sustained improvement across the evaluation period. The lag-decay diagnostic shows predictive power peaking at contemporaneous and lag-1 specifications and decaying at the 6–12-month lags. Cross-commodity application on Arabica coffee and raw sugar does not reproduce the cocoa result. Additional sensitivity tests replacing the continuous Niño 3.4 index with rolling-standard-deviation variants (12- and 36-month windows) yield the same crisis-concentrated pattern. The interpretation of this combined pattern, and what it implies for the answer to RQ2, is taken up in Section 6.2.

5.6 Summary of Findings

Table 11 consolidates the key metrics for all models across the horizon spectrum. Three conclusions emerge.

Table 11: Consolidated model comparison across horizons

Metric	Model	Horizon				
		1d	1m	3m	6m	12m
QLIKE	Historical Vol	−7.156	−7.420*	−7.405	−7.380	−7.353
	GARCH(1,1)	−7.239	−7.401*	−7.375	−7.348	−7.336
	HAR-OLS	**−7.291**	**−7.493***	**−7.474**	**−7.438**	**−7.423**
QL ($\tau = 0.95$)	Historical Vol	0.001282	0.000769*	0.000803	0.000843	0.000862
	GARCH(1,1)	0.001307	0.000905	0.000956	0.000971	0.000932*
	HAR-OLS	0.001228	0.000509	0.000493*	0.000513	0.000493*
	HAR-QR	**0.001160**	**0.000470**	0.000461*	0.000510	0.000516
	HAR-X-QR	0.001161	0.000476	**0.000442**	**0.000419**	**0.000357***

Bold: best model at each horizon. *Best horizon for each model. QLIKE: more negative is better; QL: lower is better. 1w horizon omitted for compactness.

First, RQ1 is answered affirmatively. HAR-based quantile regression outperforms both naive benchmarks at every horizon: historical volatility and GARCH(1,1) are excluded from $\mathcal{M}^*$ at every horizon under the upper-tail quantile loss criterion (Table 8), confirming that their inferior performance is not a product of sampling variation. Against the HAR-OLS pseudo-quantile, direct quantile estimation wins clearly at short horizons (HAR-QR holds $p = 1.00$ through one month) and the two are statistically indistinguishable at medium and long horizons, where both remain in $\mathcal{M}^*$. The slight numerical edge of HAR-OLS at twelve months therefore does not constitute a statistically significant defeat. From three months onward, HAR-X-QR takes the top MCS rank, providing the cleaner answer to RQ1 at long horizons. The QLIKE ranking is robust: HAR-OLS dominates mean forecasts at all horizons, and the HAR family consistently separates from GARCH under MSE and MAE as well (Figure 4).

Second, regarding RQ2, the Niño 3.4 index produces a substantial improvement at long horizons over the full evaluation sample: the $\tau = 0.95$ quantile loss falls by 30.86% at twelve months (Table 6), and HAR-X-QR holds the top MCS rank from three months onward (Table 8). Robustness testing in Section 5.5 shows that the cumulative gain is concentrated in the 2023–2024 portion of the evaluation period, follows a lag structure inconsistent with the agronomic-transmission hypothesis, and does not reproduce on comparable tropical commodities. The interpretation of this combined pattern is developed in Section 6.2.

Third, coverage analysis reveals a systematic challenge at long horizons. All models produce violation rates above the nominal 5% at six and twelve months. ENSO augmentation introduces a recalibration effect at the upper tail, raising the violation rate further while lowering quantile loss. The implications of this calibration pattern are discussed in Section 6.4.

6 Discussion

The empirical results in Section 5 establish two findings that require interpretation. The HAR family dominates both the non-parametric and the GARCH benchmarks at every horizon and across every loss criterion examined (RQ1), and the Niño 3.4 climate index produces a numerically large improvement at long horizons over the full evaluation sample, but one that does not survive the three robustness diagnostics applied to it (RQ2). This section explains why these patterns arise, how they relate to the existing literature on volatility forecasting and ENSO–commodity linkages, and what they imply for the calibration of distributional forecasts, the procurement perspective that motivated the choice of cocoa, and the directions for further research.

6.1 Why the HAR Family Dominates

The dominance of the HAR family over GARCH(1,1) and the rolling historical benchmark is structural rather than incidental. Corsi (2009) shows that standard short-memory volatility models, including GARCH, fail to reproduce the scaling behaviour of volatility under temporal aggregation: their exponential memory decay erases shocks too rapidly, and multi-step forecasts converge toward the unconditional variance within a few weeks. The HAR structure approximates the hyperbolic decay that empirical volatility exhibits by stacking heterogeneous lags at daily, weekly, and monthly frequencies, without imposing a fractional integration parameter. This matters with particular force for cocoa, which Alfeus and Nikitopoulos (2022) identify as a long-memory commodity in which monthly volatility carries the most predictive weight. Section 5.1 reports a correlation of 0.66 between the monthly HAR component and the twelve-month volatility target, against 0.37 for the daily component, which is the direct empirical counterpart of Corsi's scaling argument and explains why HAR-OLS retains its lead even at twelve months while GARCH(1,1) flatlines.

Within the HAR family, the comparison between HAR-QR and the HAR-OLS pseudo-quantile addresses a more subtle question: does the conditional upper quantile diverge sufficiently from a shift of the conditional mean to justify the additional estimation cost of quantile regression? Haugom et al. (2016) argue that the gain comes from two sources: direct quantile estimation is robust to distributional misspecification, and the monthly volatility component exerts its strongest influence precisely at the conditional tails. The cocoa results match this mechanism. The HAR-QR advantage over the HAR-OLS pseudo-quantile is largest at the short to medium horizons (10 % at one month, 6 % at one day), where volatility dynamics are most state-dependent and the conditional quantile is furthest from the unconditional residual quantile, and narrows to less than 2 % at twelve months once the HAR regressors themselves carry most of the distributional information. Lyócsa and Stašek (2021) report the same horizon dependence on stock-market data: the value of explicit quantile estimation is concentrated at short horizons where extreme regime states drive the conditional distribution. Coverage at the one-day and one-week horizons is close to the nominal 5 % (5.17 % and 5.30 %), which replicates Haugom's finding that HAR-QR passes conditional-coverage tests at substantially higher rates than historical simulation, now extended from equity to agricultural commodity data.

6.2 Interpreting the ENSO Signal

The ENSO result is harder to read than the HAR result, because the same data admit two qualitatively different interpretations. Over the full evaluation sample, HAR-X-QR achieves a 30.8 % reduction in upper-tail quantile loss at the twelve-month horizon and holds the top MCS rank from three months onward. Taken in isolation this is a large effect. The robustness diagnostics in Section 5.5 show that it is also a fragile one: the cumulative gain accumulates almost entirely within the 2022–2024 supply-crisis window, the lag structure is opposite to what an agronomic transmission chain would predict, and the gain does not replicate on Arabica coffee or raw sugar. The interpretive task is to explain how all three diagnostics can fail simultaneously while the in-sample fit remains so striking.

The physical channel that motivated the inclusion of ENSO in the first place is plausibly real but is unlikely to be visible at the lag structure assumed by the standard climate-yield literature. Schroth et al. (2016) document that severe El Niño events in the 1980s coincided with drought years that materially affected West African cocoa yields, which suggests a transmission window of six to twelve months from sea-surface anomaly to harvest impact. Our lag-decay diagnostic shows the opposite pattern: predictive power peaks at lag 1 and decays sharply by lag 6. Bergander (2025) provide the missing link: futures prices in their analysis reflect roughly 85 % of the eventual price impact of a drought shock *before* the actual crop loss occurs. If professional traders anticipate ENSO-induced supply disruption and hedge forward accordingly, the volatility footprint of an ENSO anomaly should appear near-contemporaneously with the climate signal, not lagged by the agronomic timescale. The lag-decay pattern is therefore consistent with market anticipation collapsing the physical lag, rather than with the absence of an underlying mechanism.

What anticipation alone cannot explain, however, is the crisis-concentrated character of the cumulative gain. A stable anticipation channel should deliver a steady forecast improvement across decades of ENSO cycles, not a gain that materialises in two years and is invisible before them. The plausible bridge between these two observations is the rare-disaster reading proposed by Bonato et al. (2023): ENSO acts as a proxy for the level of uncertainty over future supply, and risk-averse inventory holders respond to high ENSO uncertainty by building stocks, which in turn raises the convenience yield and amplifies the variance of returns. Under this reading, ENSO is predictive of upper-tail volatility only when the underlying supply system is already close to a stress threshold; in quiet periods, the same Niño 3.4 readings carry no incremental information beyond the HAR lags. The 2022–2024 episode supplies precisely that stress threshold via the CSSVD outbreak and the resulting structural supply shortfall, and the ENSO regressor identifies the El Niño transition that coincides with it. Bouri et al. (2021) find a structurally similar pattern for oil, where the predictive value of ENSO is concentrated at specific horizons (two to four years) and emerges most strongly under longer estimation windows that capture full climate cycles; the horizon over which the signal becomes visible differs across markets, but the regime-conditional character of the effect is shared.

This reading is also consistent with the cross-commodity null. Ubilava (2018) reports that ENSO produces in-sample evidence of predictability for selected non-agricultural commodities but that the out-of-sample relationship typically fails to replicate, and that the effect is most amplified for cocoa during El Niño episodes specifically. The HAR-X-QR results extend this pattern: coffee and sugar share the same equatorial belt and the same Pacific climate exposure, yet the same regressor that produces a 31 % improvement for cocoa *worsens* long-horizon coffee forecasts and produces a gain on sugar that is a factor of thirty-two smaller than on cocoa. A structural climate channel operating uniformly across tropical soft commodities is hard to reconcile with this asymmetry; a regime-state indicator that happens to align with one severe cocoa-specific episode is not. Huang et al. (2022) provide further support for this reading by showing that HAR coefficients differ substantially across market regimes; the ENSO coefficient, on this interpretation, is identified predominantly in the regime where its sign and magnitude happen to align with the realised volatility surprise.

The data therefore support a cautious answer to RQ2. ENSO improves upper-tail cocoa volatility forecasts in the in-sample dimension; the improvement is large and statistically significant on the full sample; and a coherent mechanism, combining market anticipation with a rare-disaster amplification through inventory dynamics, can rationalise both why the predictive power is near-contemporaneous and why it concentrates in a single crisis. None of these facts establishes that the same regressor will deliver comparable performance in future ENSO cycles. The honest reading is that HAR-X-QR captures a sample-specific covariation between the ENSO state and the cocoa supply crisis, rather than a replicable structural climate channel.

6.3 Forecast Instability and the Crisis Episode

The pattern just described is not an isolated peculiarity of cocoa. It is a textbook instance of the forecast-instability phenomenon documented in the financial-econometrics literature. Paye and Timmermann (2006) show that the discrepancy between strong in-sample predictability and weak out-of-sample performance, repeatedly observed across return-prediction studies, is best explained by structural instability in the underlying predictive relationship: predictability is real but concentrated in identifiable subperiods rather than uniform over time. Rossi (2020) argues that instability is the rule in macroeconomic and financial forecasting, not the exception, and that diagnostic procedures aimed at locating the periods in which predictability is concentrated should now be considered part of standard practice.

The expanding-window diagnostic in Section 5.5 is designed to operate within exactly that tradition. It imposes no break date, partitions no sample ex post, and traces the cumulative loss difference recursively from the first feasible forecast onward. The finding that the cumulative ENSO gain at the twelve-month upper tail is indistinguishable from zero through three decades of out-of-sample evaluation and then accumulates the entire full-sample improvement within two years is diagnostically clean: the partition emerges from the data, not from the researcher. Poon and Granger (2003) warn that in-sample parameter stability is no guarantee of out-of-sample stability, and that time variation in parameter estimates is the critical issue in forecast evaluation; the expanding-window construction addresses this warning directly.

The Model Confidence Set complements rather than contradicts this picture. Hansen et al. (2011) stress that the MCS acknowledges the limitations of the data: informative samples yield small confidence sets, uninformative samples include many models. The HAR-X-QR $p = 1.00$ rank from three months onward is therefore a full-sample statement about average predictive ability; it cannot, by construction, separate a stable channel from a single episode that dominates the average loss. Read together, the MCS result and the expanding-window diagnostic are entirely consistent: the former says that, averaged over the full sample, HAR-X-QR is statistically indistinguishable from the best model; the latter says that the average is shaped by one specific subperiod.

A brief note on a specification that was tested but not retained in the main analysis. An earlier robustness exercise included a hand-coded 2022–2024 supply-crisis dummy, which produced the visually strongest long-horizon ENSO improvement (approximately 36 % at twelve months) and the cleanest within-period attribution of the gain to the crisis window. The construction of the indicator, however, relied on press-derived boundary dates rather than a reproducible statistical rule, and the resulting variable is observationally indistinguishable from any alternative narrative for the same window: the Russian invasion of Ukraine, the post-pandemic liquidity expansion, or the compounding effects of CSSVD and black-pod infections would produce an identical dummy. The specification was therefore dropped on identification grounds. The formally estimated supremum-Wald break date reported in Section 4 provides the objectifiable counterpart; a fully developed structural-break treatment of the cocoa volatility series, in the spirit of Bai and Perron (1998), is left to future work.

6.4 Coverage, Calibration, and the Discrimination Tradeoff

Two coverage results in Section 5.2 appear, on a first reading, to be in tension with the quantile-loss rankings. The HAR-QR violation rate at the twelve-month upper quantile rises to 10.36 % against a nominal 5 %, and the HAR-X-QR violation rate rises further to 12.66 %, yet HAR-X-QR achieves a substantially lower quantile loss than HAR-QR. Christoffersen (1998) provides the relevant distinction: unconditional coverage and conditional coverage are separate properties, and a forecast can satisfy the former while violating the latter through clustered exceedances. The cocoa violations at long horizons are almost certainly clustered in the 2022–2024 episode, so the failure of unconditional coverage is, if anything, a conservative summary of the calibration problem at the upper tail.

The apparent contradiction between higher violations and lower quantile loss is resolved by recognising that the two metrics measure orthogonal properties of a distributional forecast. Dimitriadis and Puke (2026) formalise this distinction in a score-decomposition framework: average loss under a strictly consistent scoring function rewards *discrimination*, the ability to distinguish high-risk from low-risk states, whereas backtest hit rates assess *calibration*, the ability to match nominal coverage. The two are not jointly maximised. The ENSO coefficient in HAR-X-QR raises the upper-quantile forecast during El Niño episodes, which reduces the magnitude of large exceedances when realised volatility spikes, and lowers it during La Niña, which produces more small exceedances in otherwise calm periods. The asymmetric check function penalises a small number of large under-predictions more heavily than a larger number of small ones, so the model that re-allocates its boundary toward the conditions that produce the most damaging exceedances wins on quantile loss while losing on average violation rate. HAR-X-QR is, in this specific sense, better calibrated where it matters most for distributional ranking and less well calibrated overall.

Patton (2011) provides a complementary argument for prioritising the score over the hit rate in this setting. Among the loss functions commonly used in volatility evaluation, only MSE and QLIKE are robust to noise in the volatility proxy; coverage diagnostics compare a binary indicator against the noisy proxy directly, and their informativeness deteriorates as proxy noise rises with the forecast horizon. The Yang-Zhang proxy becomes noisier as the averaging horizon extends; the quantile-loss ranking is, on

Patton's argument, more trustworthy at twelve months than the violation-rate comparison. Which property a user should prioritise depends on the use case: distributional ranking favours the score, nominal-coverage compliance favours the violation rate, and the two should be reported together.

6.5 Procurement Outlook

The thesis answers a methodological question rather than a procurement question, and the formal evaluation of a hedging strategy lies outside its scope. The results nonetheless connect naturally to the procurement perspective that motivated the choice of cocoa as the test market. Poon and Granger (2003) conclude their survey of the volatility-forecasting literature by recommending simple historical methods for horizons longer than six months, on the grounds that more elaborate models fail to deliver consistent gains at those horizons. The cocoa results suggest that this recommendation, at least for distributional forecasting of an agricultural commodity, may be conservative: HAR-QR reduces the upper-tail quantile loss against historical volatility by 67 % at the twelve-month horizon, and HAR-X-QR by 142 %.

A mechanism that links these distributional forecasts to the operational behaviour of physical-cocoa users is provided by Gilbert and Morgan (2010), who show that stockholding causes volatility to bunch: small shocks have outsized price impacts when inventories are low, and the same shocks have muted impacts when stocks are high. The HAR-X-QR boundary, by rising during El Niño episodes that correlate with periods of inventory tightness, traces precisely the conditions in which Gilbert and Morgan's amplification effect should be expected to bind. The direct projection design avoids the error accumulation that affects iterated multi-period forecasts (Ghysels and Marcellino 2018), and the coherence of the check function as both the estimation objective and the evaluation metric (Puke and Schweikert 2026) is preserved across all horizons reported. Whether a procurement strategy built on these forecasts delivers measurable economic value once transaction costs, basis risk, and asymmetric loss structures are accounted for is a separate question that a properly designed strategy evaluation would have to answer.

6.6 Limitations and Future Research

Six limitations of the present specification deserve explicit mention. First, the HAR-QR coefficients are static within each estimation window. Gong et al. (2017) show that a HAR specification with time-varying sparsity, using normal-gamma autoregressive priors, substantially outperforms standard HAR for agricultural commodity futures; a time-varying HAR-QR with smoothly evolving quantile coefficients is a natural next step and would be expected to reduce the coverage degradation observed at long horizons. Second, the model has no explicit regime structure. Luo et al. (2022) demonstrate that infinite-Hidden-Markov regime switching delivers substantial gains over benchmark HAR for agricultural commodities precisely when macroeconomic uncertainty is high; the 2022–2024 episode is exactly the kind of high-uncertainty regime that an IHM-HAR-QR would be designed to model directly. Third, the model conditions on volatility lags but not on realised jumps, skewness, or higher moments; Andersen et al. (2007) and Bonato et al. (2024) both show that these components carry forecast-relevant information beyond what HAR lags capture, and the upper-tail coverage failures at long horizons are a plausible signature of unmodelled jumps in the crisis period.

Fourth, the climate exposure is summarised by a single index. The Indian Ocean Dipole and the Atlantic Meridional Mode also influence West African rainfall patterns and could disentangle Pacific from Atlantic teleconnections in a multi-index extension. Fifth, the analysis uses ICE London front-month rolling futures denominated in GBP only; ICE New York contracts, basis dynamics between the two markets, and contract-structure effects are not modelled, and a comparative two-exchange analysis would provide a natural robustness extension. Sixth, the estimation window length is fixed at 2,520 trading days. Bouri et al. (2021) report that the predictive value of ENSO on oil volatility emerges most strongly under estimation windows of four to six years; the ten-year window used here may dilute regime-specific coefficients and dampen climate-driven episodes. An adaptive-window design that lets the data choose the relevant historical horizon for each forecast is a methodological extension worth pursuing.

Beyond these direct extensions, the broader research programme suggested by the cocoa results points in three directions. A jump-augmented HAR-J-QR specification would address the upper-tail coverage problem at the source. A regime-switching IHM-HAR-QR would model directly what the present specification absorbs through a single ENSO coefficient. A multi-predictor extension combining the Niño 3.4 index with realised inventory data, shipping rates, or other proxies of physical supply tightness would test whether the ENSO gain identified here can be disentangled from the broader regime indicators that coincide with it. Each of these directions would require both the data infrastructure and the methodological development that lie outside the scope of a single master's thesis; together they define the natural research agenda that follows from the findings reported in Section 5.

The overall picture is consistent. HAR-based quantile regression provides a substantial and statistically significant improvement over the standard volatility-forecasting benchmarks for cocoa at every horizon examined, with the upper-tail advantage strongest at the longest horizons that lie far beyond the prior literature limit. The ENSO augmentation produces a numerically striking long-horizon gain that, on close examination, reflects a single severe episode rather than a replicable climate channel. The combination of strong RQ1 evidence and a carefully delimited RQ2 result positions the thesis as a methodological contribution to the volatility-forecasting literature on agricultural commodities, while flagging precisely where the limits of what daily-frequency data and a single decade of evaluation can reveal about a slow-moving climate signal in fact lie.

References

Alfeus, Mesias and Christina Sklibosios Nikitopoulos (2022). “Forecasting Volatility in Commodity Markets with Long-Memory Models”. In: *Journal of Commodity Markets* 28, p. 100248. DOI: 10.1016/j.jcomm.2022.100248.

Andersen, Torben G. and Tim Bollerslev (1998). “Answering the Skeptics: Yes, Standard Volatility Models Do Provide Accurate Forecasts”. In: *International Economic Review* 39.4, pp. 885–905. DOI: 10.2307/2527343.

Andersen, Torben G., Tim Bollerslev, and Francis X. Diebold (2007). “Roughing It Up: Including Jump Components in the Measurement, Modeling, and Forecasting of Return Volatility”. In: *Review of Economics and Statistics* 89.4, pp. 701–720. DOI: 10.1162/rest.89.4.701.

Andersen, Torben G., Tim Bollerslev, Francis X. Diebold, and Paul Labys (2001). “The Distribution of Realized Exchange Rate Volatility”. In: *Journal of the American Statistical Association* 96.453, pp. 42–55. DOI: 10.1198/016214501750332965.

Asante, Paulina A., Eric Rahn, Niels P. R. Anten, Pieter A. Zuidema, Alejandro Morales, and Danaë M. A. Rozendaal (2025). “Climate change impacts on cocoa production in the major producing countries of West and Central Africa by mid-century”. In: *Agricultural and Forest Meteorology*. DOI: 10.1016/j.agrformet.2025.110339.

Bai, Jushan and Pierre Perron (1998). “Estimating and Testing Linear Models with Multiple Structural Changes”. In: *Econometrica* 66.1, pp. 47–78. DOI: 10.2307/2998540.

Baur, Dirk G. and Thomas Dimpfl (2019). “A Quantile Regression Approach to Estimate the Variance of Financial Returns”. In: *Journal of Financial Econometrics* 17.4, pp. 616–644. DOI: 10.1093/jjfinec/nby027.

Bensch, Gunther, Kathrin Kaestner, Laura Schmid, Colin Vance, and Adalbert Wilhelm (2022). *Final Report for the Federal Agency for Agriculture and Food: Price Formation on the Cocoa Futures Market and on Physical Markets for Raw Cocoa*. Research Report. RWI – Leibniz-Institut für Wirtschaftsforschung.

Bergander, Felix (2025). “Deep Learning for Cocoa Price Forecasting: Integrating Weather and Fundamental Data”. In: *Working Paper*.

Bollerslev, Tim (1986). “Generalized Autoregressive Conditional Heteroskedasticity”. In: *Journal of Econometrics* 31.3, pp. 307–327. DOI: 10.1016/0304-4076(86)90063-1.

Bonato, Matteo, Oğuzhan Çepni, Rangan Gupta, and Christian Pierdzioch (2023). “El Niño, La Niña, and Forecastability of the Realized Variance of Agricultural Commodity Prices: Evidence from a Machine Learning Approach”. In: *Journal of Forecasting* 42.4, pp. 785–801. DOI: 10.1002/for.2914.

Bonato, Matteo, Oğuzhan Çepni, Rangan Gupta, and Christian Pierdzioch (2024). "Forecasting the Realized Volatility of Agricultural Commodity Prices: Does Sentiment Matter?" In: *Journal of Forecasting* 43.6, pp. 2088–2125. DOI: 10.1002/for.3106.

Bouri, Elie, Rangan Gupta, Christian Pierdzioch, and Afees A. Salisu (2021). "El Niño and Forecastability of Oil-Price Realized Volatility". In: *Theoretical and Applied Climatology* 144, pp. 1173–1180. DOI: 10.1007/s00704-021-03569-1.

Christoffersen, Peter F. (1998). "Evaluating Interval Forecasts". In: *International Economic Review* 39.4, pp. 841–862. DOI: 10.2307/2527341.

Clements, Adam, Stan Hurn, and Vladimir Volkov (2024). "Harvesting HAR-X Volatility Forecasts Using OHLC Data". In: *Journal of Empirical Finance* 79, p. 101555. DOI: 10.1016/j.jempfin.2024.101555.

Corsi, Fulvio (2009). "A Simple Approximate Long-Memory Model of Realized Volatility". In: *Journal of Financial Econometrics* 7.2, pp. 174–196. DOI: 10.1093/jjfinec/nbp001.

Degiannakis, Stavros, George Filis, Tony Klein, and Thomas Walther (2022). "Forecasting Realized Volatility of Agricultural Commodities". In: *International Journal of Forecasting* 38.1, pp. 74–96. DOI: 10.1016/j.ijforecast.2019.08.011.

Dickey, David A. and Wayne A. Fuller (1979). "Distribution of the Estimators for Autoregressive Time Series with a Unit Root". In: *Journal of the American Statistical Association* 74.366, pp. 427–431. DOI: 10.2307/2286348.

Dimitriadis, Timo and Marius Puke (2026). "Statistical Inference for Score-Based Forecast Evaluation". In: *Working Paper*.

Engle, Robert F. (1982). "Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation". In: *Econometrica* 50.4, pp. 987–1007. DOI: 10.2307/1912773.

Ge, Yiqun, Xinru Chen, and Lili Ding (2025). "Volatility in a Mug Cup: Forecasting and Spillover Analysis of Cocoa, Coffee and Sugar Futures". In: *Finance Research Letters* 75, p. 106810. DOI: 10.1016/j.frl.2025.106810.

Ghysels, Eric and Massimiliano Marcellino (2018). "Applied Economic Forecasting Using Time Series Methods". In.

Gilbert, Christopher L. and C. Wyn Morgan (2010). "Food Price Volatility". In: *Philosophical Transactions of the Royal Society B: Biological Sciences* 365.1554, pp. 3023–3034. DOI: 10.1098/rstb.2010.0139.

Gong, Xu, Zengyan Si, Ping Jiang, and Qi Ma (2017). "Realized Volatility Forecasting of Agricultural Commodity Futures Using the HAR Model with Time-Varying Sparsity". In: *International Journal of Forecasting* 33.1, pp. 132–152. DOI: 10.1016/j.ijforecast.2016.09.004.

Guo, Yangli, Pei Peng, Ling Zhou, and Yusui Tang (2025). “Forecasting Volatility in Commodity Markets with Climate Risk”. In: *Finance Research Letters* 71, p. 106357. DOI: 10.1016/j.frl.2024.106357.

Hansen, Peter R. and Asger Lunde (2005). “A Forecast Comparison of Volatility Models: Does Anything Beat a GARCH(1,1)?” In: *Journal of Applied Econometrics* 20.7, pp. 873–889. DOI: 10.1002/jae.800.

Hansen, Peter R., Asger Lunde, and James M. Nason (2011). “The Model Confidence Set”. In: *Econometrica* 79.2, pp. 453–497. DOI: 10.3982/ECTA5771.

Haugom, Erik, Rina Ray, Carl J. Ullrich, Steinar Veka, and Sjur Westgaard (2016). “A Parsimonious Quantile Regression Model to Forecast Day-Ahead Value-at-Risk”. In: *Finance Research Letters* 16, pp. 196–207. DOI: 10.1016/j.frl.2015.12.006.

Huang, Jiefei, Yang Xu, and Yuping Song (2022). “A High-Frequency Approach to VaR Measures and Forecasts Based on the HAR-QREG Model with Jumps”. In: *Physica A: Statistical Mechanics and its Applications* 608, p. 128253. DOI: 10.1016/j.physa.2022.128253.

Institute, International Food Policy Research (2024). “Soaring Cocoa Prices: Diverse Impacts and Implications for Key West African Producers”. In: *IFPRI Blog*. Accessed 2026-05-29.

Koenker, Roger and Gilbert Bassett (1978). “Regression Quantiles”. In: *Econometrica* 46.1, pp. 33–50. DOI: 10.2307/1913643.

Koenker, Roger and Vasco d’Orey (1987). “Algorithm AS 229: Computing Regression Quantiles”. In: *Applied Statistics* 36.3, pp. 383–393. DOI: 10.2307/2347802.

Läderach, Peter, Armando Martinez-Valle, Götz Schroth, and Nestor Castro (2013). “Predicting the future climatic suitability for cocoa farming of the world’s leading producer countries, Ghana and Côte d’Ivoire”. In: *Climatic Change* 119.3–4, pp. 841–854. DOI: 10.1007/s10584-013-0774-8.

Lindsey, Rebecca (2009). *Climate Variability: Oceanic Niño Index*.

Luo, Jiawen, Tony Klein, Qiang Ji, and Chenghan Hou (2022). “Forecasting Realized Volatility of Agricultural Commodity Futures with Infinite Hidden Markov HAR Models”. In: *International Journal of Forecasting* 38.1, pp. 51–73. DOI: 10.1016/j.ijforecast.2019.08.008.

Lyócsa, Štefan and Daniel Stašek (2021). “Improving Stock Market Volatility Forecasts with Complete Subset Linear and Quantile HAR Models”. In: *Expert Systems with Applications* 183, p. 115370. DOI: 10.1016/j.eswa.2021.115370.

Müller, Ulrich A., Michel M. Dacorogna, Rakhal D. Davé, Richard B. Olsen, Olivier V. Pictet, and Jakob E. von Weizsäcker (1997). “Volatilities of Different Time Resolutions – Analyzing the Dynamics of Market Components”. In: *Journal of Empirical Finance* 4.2-3, pp. 213–239. DOI: 10.1016/S0927-5398(97)00007-8.

Organization, International Cocoa (2024). “Cocoa Market Report for March 2024”. In: *ICCO Market Report*. Accessed 2026-05-29.

Parkinson, Michael (1980). “The Extreme Value Method for Estimating the Variance of the Rate of Return”. In: *Journal of Business* 53.1, pp. 61–65. DOI: 10.1086/296071.

Patton, Andrew J. (2011). “Volatility Forecast Comparison Using Imperfect Volatility Proxies”. In: *Journal of Econometrics* 160.1, pp. 246–256. DOI: 10.1016/j.jeconom.2010.03.034.

Patton, Andrew J. and Kevin Sheppard (2015). “Good Volatility, Bad Volatility: Signed Jumps and the Persistence of Volatility”. In: *Review of Economics and Statistics* 97.3, pp. 683–697. DOI: 10.1162/REST_a_00503.

Paye, Bradley S. and Allan Timmermann (2006). “Instability of Return Prediction Models”. In: *Journal of Empirical Finance* 13.3, pp. 274–315. DOI: 10.1016/j.jempfin.2005.11.003.

Phillips, Peter C. B. and Pierre Perron (1988). “Testing for a Unit Root in Time Series Regression”. In: *Biometrika* 75.2, pp. 335–346. DOI: 10.1093/biomet/75.2.335.

Poon, Ser-Huang and Clive W. J. Granger (2003). “Forecasting Volatility in Financial Markets: A Review”. In: *Journal of Economic Literature* 41.2, pp. 478–539. DOI: 10.1257/002205103765762743.

Puke, Marius and Karsten Schweikert (2026). “Coherent Forecasting of Realized Volatility”. In: *Journal of Forecasting*. DOI: 10.1002/for.70114.

Rogers, L. C. G. and S. E. Satchell (1991). “Estimating Variance from High, Low, and Closing Prices”. In: *Annals of Applied Probability* 1.4, pp. 504–512. DOI: 10.1214/aoap/1177005835.

Rogna, Marco and Pierre Tillie (2025). “An Analysis of Cocoa Market Fundamentals and Price Transmission”. In: *Journal of Commodity Markets* 37, p. 100421. DOI: 10.1016/j.jcomm.2024.100421.

Rossi, Barbara (2020). *Forecasting in the Presence of Instabilities: How Do We Know Whether Models Predict Well and How to Improve Them*. Tech. rep. Barcelona Graduate School of Economics.

Schroth, Götz, Peter Läderach, Armando Isaac Martinez-Valle, Christian Bunn, and Laurence Jassogne (2016). “Vulnerability to climate change of cocoa in West Africa: Patterns, opportunities and limits to adaptation”. In: *Science of the Total Environment* 556, pp. 231–241. DOI: 10.1016/j.scitotenv.2016.03.024.

Song, Yan, Jiahui Pei, Yufeng Li, and Yuping Song (2025). “Gold Futures Volatility Forecasting Based on HAR-QR and Machine Learning Model”. In: *Artificial Intelligence and Human-Computer Interaction*. IOS Press. DOI: 10.3233/FAIA250131.

Tian, Fengping, Ke Yang, and Langnan Chen (2017). “Realized Volatility Forecasting of Agricultural Commodity Futures Using the HAR Model with Time-Varying Sparsity". In: *International Journal of Forecasting* 33.1, pp. 132–152. DOI: 10.1016/j.ijforecast.2016.08.002.

Tsay, Ruey S. (2010). *Analysis of Financial Time Series*. 3rd. John Wiley & Sons.

Ubilava, David (2018). "The Role of El Niño Southern Oscillation in Commodity Price Movement and Predictability". In: *American Journal of Agricultural Economics* 100.1, pp. 239–263. DOI: 10.1093/ajae/aax060.

Yang, Dennis and Qiang Zhang (2000). "Drift-Independent Volatility Estimation Based on High, Low, Open, and Close Prices". In: *Journal of Business* 73.3, pp. 477–491. DOI: 10.1086/209650.

Appendices

A Supplementary Tables

This appendix collects reference material and detailed estimates that support the main text but are not essential to the line of argument: the data description (Table 12), the unit root tests (Table 13), the Mincer-Zarnowitz validation of the Yang-Zhang proxy against intraday realized volatility (Table 14), and the full coefficient estimates of the HAR-QR and HAR-X-QR models (Table 15).

Table 12: Summary of Cocoa Futures Data

Characteristic	**Value**
Exchange	ICE Futures Europe
Contract	LCC (Cocoa Futures)
Contract size	10 metric tonnes
Price quotation	GBP per metric tonne
Continuous series	LCCc1 (front-month rolling)
Data source	Refinitiv Workspace
Sample period (daily)	January 2, 1985 – December 15, 2025
Number of observations (daily)	10,376 trading days
Sample period (5-minute)	February 3, 2025 – January 12, 2026
Number of observations (5-minute)	20,791 intraday observations
Variables (daily)	Open, High, Low, Close, Volume
Variables (5-minute)	Open, High, Low, Close, Volume, Bid, Ask

Table 13: Unit root tests for stationarity

		ADF		PP	
Series	N	Statistic	p-value	Statistic	p-value
Log returns	10,375	-72.06	< 0.001	-98.89	< 0.001
Yang-Zhang volatility	10,375	-6.27	< 0.001	-77.58	< 0.001
Niño 3.4 anomaly	787	-6.85	< 0.001	-6.46	< 0.001

ADF = Augmented Dickey–Fuller test (Dickey and Fuller 1979); PP = Phillips–Perron test (Phillips and Perron 1988). Both tests include a constant. ADF lag length selected by AIC. The null hypothesis is a unit root; rejection indicates stationarity.

Table 14: Mincer-Zarnowitz regression: Yang-Zhang against intraday realized volatility

	$\hat{\alpha}$	$\hat{\beta}$	R^2	N	$F_{\alpha=0,\,\beta=1}$	p-value
Daily	0.0127***	0.4589***	0.464	196	75.1	< 0.001
	(0.0011)	(0.0443)				
22-day rolling	0.0022***	0.8355***	0.847	175	140.9	< 0.001
	(0.0006)	(0.0247)				

Mincer-Zarnowitz regression $\mathrm{RV}_t = \alpha + \beta\,\hat{\sigma}_t^{\mathrm{YZ}} + \varepsilon_t$ estimated with HC1 standard errors (in parentheses). The F-statistic tests the joint hypothesis $\alpha = 0$, $\beta = 1$ (unbiased proxy). RV is computed from five-minute log returns; $\hat{\sigma}^{\mathrm{YZ}}$ is the Yang-Zhang estimator from daily OHLC prices. The 22-day row uses rolling 22-day averages of both series, aligning with the HAR monthly component. The lower $\hat{\beta}$ at the daily frequency reflects the different treatment of the overnight component: Yang-Zhang captures the close-to-open gap while intraday RV does not. Sample: 3 February 2025 – 12 January 2026. *** $p < 0.001$.

Table 15: Full-sample HAR-QR and HAR-X-QR coefficient estimates

Horizon	Model	β_0	β_d	β_w	β_m	β_{ENSO}
Panel: $\tau = 0.50$						
1 day	HAR-QR	0.0014	0.1231	0.1721	0.4950	—
	HAR-X-QR	0.0015	0.1195	0.1732	0.4900	-0.0001
1 week	HAR-QR	0.0023	0.0729	0.1317	0.5745	—
	HAR-X-QR	0.0024	0.0717	0.1341	0.5671	-0.0001
1 month	HAR-QR	0.0034	0.0329	0.1264	0.5669	—
	HAR-X-QR	0.0034	0.0345	0.1210	0.5650	-0.0002
3 months	HAR-QR	0.0044	0.0186	0.0684	0.5670	—
	HAR-X-QR	0.0045	0.0181	0.0718	0.5553	-0.0003
6 months	HAR-QR	0.0060	0.0199	0.0489	0.4850	—
	HAR-X-QR	0.0061	0.0199	0.0442	0.4761	-0.0004
12 months	HAR-QR	0.0078	0.0144	0.0400	0.3626	—
	HAR-X-QR	0.0078	0.0131	0.0399	0.3606	-0.0002
Panel: $\tau = 0.75$						
1 day	HAR-QR	0.0026	0.1742	0.2477	0.5819	—
	HAR-X-QR	0.0027	0.1772	0.2471	0.5689	-0.0001
1 week	HAR-QR	0.0026	0.0975	0.1989	0.6606	—
	HAR-X-QR	0.0028	0.0985	0.1934	0.6566	-0.0000
1 month	HAR-QR	0.0039	0.0507	0.1402	0.6533	—
	HAR-X-QR	0.0039	0.0468	0.1341	0.6599	-0.0002
3 months	HAR-QR	0.0054	0.0296	0.0559	0.6377	—
	HAR-X-QR	0.0055	0.0280	0.0547	0.6317	-0.0003
6 months	HAR-QR	0.0070	0.0209	0.0463	0.5485	—
	HAR-X-QR	0.0070	0.0192	0.0488	0.5419	-0.0003
12 months	HAR-QR	0.0090	0.0133	0.0333	0.4250	—
	HAR-X-QR	0.0090	0.0136	0.0334	0.4249	-0.0000
Panel: $\tau = 0.95$						
1 day	HAR-QR	0.0055	0.3740	0.4437	0.7203	—
	HAR-X-QR	0.0055	0.3787	0.4366	0.7209	-0.0002
1 week	HAR-QR	0.0042	0.2051	0.3034	0.7817	—
	HAR-X-QR	0.0041	0.2057	0.3074	0.7814	-0.0002
1 month	HAR-QR	0.0050	0.0559	0.1349	0.9099	—
	HAR-X-QR	0.0050	0.0520	0.1563	0.8886	-0.0001
3 months	HAR-QR	0.0068	0.0376	0.2375	0.6378	—
	HAR-X-QR	0.0068	0.0303	0.2437	0.6368	-0.0001
6 months	HAR-QR	0.0076	0.0261	0.1359	0.6769	—
	HAR-X-QR	0.0079	0.0205	0.1526	0.6602	0.0003
12 months	HAR-QR	0.0104	0.0225	0.1217	0.4857	—
	HAR-X-QR	0.0134	0.0082	0.1092	0.4131	0.0026

Quantile regression coefficients estimated once on the full sample for each horizon and quantile τ. β_0 is the intercept; β_d, β_w, β_m are the daily, weekly and monthly HAR components; β_{ENSO} is the Niño 3.4 loading (HAR-X-QR only). The table is descriptive: it reports the sign and magnitude of the loadings only. Significance tests are omitted because the forecast targets are overlapping h-day-ahead averages, which violates the independence assumption underlying the standard errors. The forecasting results in Section 5 use rolling-window re-estimation; these full-sample estimates are reported for reference only.